Talk2MeLatin

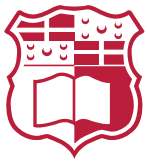
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Abstract

Artificial Intelligence (AI) has become an increasingly valuable tool in language education, offering interactive and personalised learning experiences for widely-studied modern languages. However, there remains a lack of AI-based tools that support learners of classical languages such as Latin, and existing digital resources for Latin lack the interactive capability to interpret and respond to natural-language questions. This project addresses this gap by developing a conversational chatbot system designed to answer English-language grammar questions about Latin words.

The system combines three main components: an intent classifier, a morphological analyser, and a template-based response generator, which are all accessible through a web-based conversational interface. The intent classifier is a fine-tuned BERT model trained to recognise eleven grammatical question intents, along with an additional class for when the question is out of scope. Training data for the intent classifier was generated synthetically from templates, using a Latin words dataset built by extending the UniMorph Latin dataset with additional features. The morphological analyser is a multi-head model built on top of Latin BERT, and trained on the PROIEL Treebank to predict part-of-speech and eight morphological features. A confidence-based fallback mechanism allows the system to fall back to the Latin words dataset when the analyser's confidence falls below a per-feature threshold.

The intent classifier achieved an overall accuracy and F1 score of 0.97, while the morphological analyser achieved per-feature accuracies ranging from 0.71 to 0.90, with a full-tag accuracy of 0.56. A user evaluation involving eight Latin learners revealed that, despite the high test-set performance, the system struggled to understand many real learner questions and was perceived as having limited pedagogical usefulness. The contrast between the technical and user evaluations highlighted a gap between the templated training distribution and the variety of questions actual learners ask, as well as the impact of class imbalance in the PROIEL Treebank on minority class performance.

In conclusion, this project demonstrates the feasibility of developing a conversational AI tool for Latin grammar support, while also identifying the main limitations that future work could address, including more diverse training data, larger question coverage, and improved handling of minority morphological classes.

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List of Abbreviations

AI Artificial Intelligence.

BERT Bidirectional Encoder Representations from Transformers.

ECE Expected Calibration Error.

GRU Gated Recurrent Unit.

LLM Large Language Model.

LSTM Long Short-Term Memory.

NLP Natural Language Processing.

NLTK Natural Language Toolkit.

NLU Natural Language Understanding.

POS Part of Speech.

RNN Recurrent Neural Network.

SVM Support Vector Machine.

UI User Interface.

UniMorph Universal Morphology.

1 Introduction

In recent years, the use of AI in education, particularly in language learning, has been growing [20]. The increase in AI-based features in language learning platforms such as Duolingo[1], and recently the use of Large Language Models (LLM), shows the growing recognition of AI's potential to enhance language education. These tools offer many advantages, including personalized learning experiences for the individual student's needs, as well as on-demand access to language support beyond the classroom, making them increasingly valuable in modern educational environments. Conversational interfaces and chatbots, in particular, are effective tools for practising a language as they enable learners to engage in interactive dialogue and provide instant feedback, simulating the benefits of a human tutor, and make learning more accessible for new students. The use of these tools in helping with the study of widely-spoken modern languages demonstrates how useful conversational AI in language education can be.

Despite these advances, classical languages such as Latin do not benefit from the same amount of digital resources and AI-supported educational tools that several modern languages do, creating a significant gap. Latin learners face several unique challenges. Unlike modern languages, Latin is a primarily written and academic language, with no living community of native speakers.[2] This means that students cannot rely on conversational immersion or interaction with fluent speakers as language practice. Although there are digital resources for Latin, they tend to focus on functionalities such as dictionary lookup or morphological parsing. Tools such as Whitaker's Words [38], while valuable for their morphological parsing capabilities, offer only lookup functionality and do not provide interactive explanations or assist learners in a conversational way through natural-language questions. Similarly, traditional Latin dictionaries and grammar references, although reliable, require learners to independently navigate complex grammatical concepts without interactive support.

1.1 Problem Definition

Having established the value of AI and conversational tools in language learning and the gap in such resources for Latin, the main problem this project addresses is the absence of a conversational AI system that can understand and respond to Latin grammar questions. As a result, students studying Latin lack a system that can support them when they encounter specific grammatical difficulties, and they do not have a conversational tool to ask for immediate assistance. Existing computational resources

[1] https://blog.duolingo.com/duolingo-max/

[2] https://www.discovermagazine.com/latin-is-dead-but-not-extinct-42750

for Latin function as dictionaries by providing lookup data, but they lack the interactive capability to answer specific questions.

Therefore, the specific problem this project seeks to solve can be defined as the need for an AI-based system capable of understanding user intent in natural-language grammar queries. The proposed system aims to address this by taking as input a natural-language question in English of the kind a beginner Latin learner might ask about the morphological properties of a Latin word. Unlike simple word-based lookup, learners may phrase questions in various forms, and the system must be able to interpret these and map the questions to a fixed set of grammatical concepts that may be queried. The expected output is a natural-language response that correctly identifies which grammatical aspect the question targets and provides accurate morphological information for the Latin word in question. The scope of the project is restricted to morphological queries about individual Latin words. Other questions, such as syntactic questions about full sentences, translation, or stylistic analysis, are outside the scope of this project and left for future work.

1.2 Motivation

Although Latin has no native speakers today, it is still widely studied and remains relevant in fields such as classical studies, history, law, medicine, linguistics, and theology [8]. The existence of a modern and ongoing student base with limited tools introduces the need for a tool that can make Latin more accessible to learners. Unlike students of modern languages, Latin learners cannot turn to a dedicated conversational tool for quick answers to the kinds of grammar questions that arise constantly during beginner study, and are instead left to navigate existing references on their own.

Overall, the main motivation for this project lies in the practical value of adapting modern Natural Language Processing (NLP) techniques to a low-resource and underserved language for learners. Latin receives little attention in mainstream NLP work compared to widely spoken modern languages, and the techniques developed for those languages have not been fully used for learner-facing tools for Latin. The creation of a conversational AI tool that helps with Latin grammar would therefore be a way to support Latin learners and help address the gap between classical studies and AI for language learning.

1.3 Aims and Objectives

The aim of this project is to develop a conversational tool for Latin language learners to provide answers to queries related to the morphological features of a Latin word. In

order to achieve this aim, the following objectives have been identified:

- **Objective 1 (O1)** - Construct the datasets required by the system: (i) a dataset of Latin words and their grammatical features, built by extending an existing Latin language dataset; (ii) a synthetic dataset of grammar questions annotated with intent labels for training the intent classification model. Since this project is aimed at a beginner level, these questions are formulated in English, as learners would not be able to converse in Latin as yet.
- **Objective 2 (O2)** - Fine-tune two BERT-based models; one is for intent classification to identify user queries, and a second model is a morphological analyser capable of tagging a Latin wordform with its morphological features.
- **Objective 3 (O3)** - Develop a response generation component that combines the classified intent and the morphological information to produce an appropriate natural-language answer, accessible through a conversational interface.
- **Objective 4 (O4)** - Evaluate the system through quantitative metrics on the intent classification and morphological analysis components, and qualitative user testing with Latin learners.

1.4 Document Structure

The subsequent chapters of this document are organized as follows: Chapter 2 presents the necessary background and a review of the existing literature and related work in the areas relevant to this project. Chapter 3 details the design and implementation of the system, including the architecture, dataset creation, intent classification model, morphological analysis model, and response generation. Chapter 4 presents and discusses the results obtained from the evaluation and experiments carried out. Chapter 5 summarizes the work carried out in this project, discusses its limitations, and suggests areas for future work.

2 Background and Literature Review

This chapter presents the background and literature that serves as the foundation for the system proposed in this project. It examines the role of chatbots in language learning, the available datasets and digital resources for Latin, BERT-based models, intent classification, and finally morphological analysis.

2.1 AI and Chatbots in Language Learning

AI has become increasingly integrated into language learning, as it provides personalised, adaptive, and interactive help to learners [6, 20]. For example, AI-based features have recently become more common in language learning applications. One such application is Duolingo, which uses GPT-4 to give users personalised feedback on their answers in a lesson, and to allow users to practice conversation with characters in the app.[1] Chatbots in particular are also playing an increasingly significant role in language learning as they can be used to simulate dialogue, answer questions, and provide users with explanations and feedback. Huang et al. [19] examined the usefulness of chatbots in language learning and found that they provide students with personalised learning experiences. They also found that chatbots can accommodate individual learning paces and allow students to learn at any time, since they can receive responses immediately. Belda-Medina and Calvo-Ferrer [3] similarly mention how conversational AI tools can be accessible conversational partners that are always available, especially for learners with limited opportunities to practice with native speakers. These benefits are also supported by empirical studies. Alenezi and Alenezi [1] conducted a study with secondary school students using a GPT-powered chatbot for English conversational practice, and reported significant improvements in the speaking competence of the group that interacted with the chatbot compared to a group taught through traditional methods, as well as high learner motivation, confidence, and satisfaction with the chatbot-assisted experience. The systematic review by Cislowska and Pena-Acuna [7] examines findings across many studies on the use of chatbots within foreign language education, assessing chatbots' influence on student skills, attitudes, and emotional well-being. They report that chatbots generally have a positive impact across a range of language skills, including speaking, vocabulary, and grammar, as well as on learner attitudes such as motivation and engagement, and emotional factors such as speaking anxiety and self-confidence. The review also notes some limitations, such as chatbots sometimes providing grammatical errors or nonsensical replies.

[1] `https://blog.duolingo.com/duolingo-max/`

2.2 Datasets and Resources for Latin

Choosing the right dataset is fundamental to this project as it serves as the foundation for the model itself. The Universal Morphology (UniMorph) dataset [36] provides morphologically annotated word forms for multiple languages, including Latin. The Latin portion of UniMorph[2] consists of 976,489 entries containing a lemma, an inflected word form derived from that lemma, and a set of morphological features encoded using the UniMorph feature schema. These features indicate the different grammatical categories that represent the inflected word. The UniMorph dataset offers several advantages for this project. First, its standardized annotation format facilitates the representation of features for all words, making it consistent and easy to understand. Second, it covers a large amount of Latin words. Third, its machine-readable format makes it straightforward to integrate into computational systems. For these reasons, this dataset was the main dataset chosen for this project. However, UniMorph has certain limitations in relation to this project's objectives. Most significantly, the dataset excludes specific features that are frequently requested by Latin learners seeking to understand inflectional patterns, particularly conjugation numbers for verbs and declension numbers for nouns. These gaps require strategies to expand and create a more complete dataset which are discussed in subsequent sections. Other Latin resources include syntactically annotated corpora such as the PROIEL Treebank [16]. Unlike UniMorph, which focuses on isolated words, this treebank is made up of sentences from Latin texts annotated with grammatical and syntactical information. The PROIEL Treebank consists of texts that are both Classical Latin and Late Latin.

In addition to datasets, computational tools to support Latin language learning and processing also exist. William Whitaker's Words is a morphological parsing program that analyses Latin words entered by users and returns grammatical information about inflected forms [38]. The program's output includes the lemma, English meaning, and grammatical features such as case, number, gender, tense, mood, voice, person, conjugation number, and declension number. This output format resembles the structure of the UniMorph dataset, but Whitaker's Words often provides additional features absent from UniMorph, most notably the conjugation and declension numbers. Whitaker's Words is a valuable resource for extending the grammatical dataset for this project. By automatically querying the program with word forms from UniMorph, and extracting additional feature information, a more comprehensive dataset can be constructed.

[2] https://github.com/unimorph/lat

2.3 BERT and Latin BERT

Bidirectional Encoder Representations from Transformers (BERT), introduced by Devlin et al. [11], is a language representation model that implements a Transformer encoder architecture that processes text bidirectionally by considering both left and right context simultaneously. BERT is pre-trained on unlabeled data using two unsupervised tasks, a masked language model and next sentence prediction. This pre-training enables BERT to learn general language representations. The pre-trained BERT model can then be fine-tuned for downstream tasks, such as intent classification, by adding task-specific layers and training on labelled data for that task. This transfer learning technique is very effective, particularly for low-resource tasks, because the model makes use of knowledge from pre-training, rather than having to learn entirely from scratch [11]. For this project, fine-tuning BERT for the intent classification task on Latin grammar questions will enable the model to recognise certain patterns such as question structures and grammar-related terminology.

Recognising the limitations of generic pre-trained models for Latin-specific tasks, Bamman and Burns [2] developed Latin BERT, a BERT model trained specifically on Latin texts. Latin BERT was trained on a large corpus of Latin text from several sources, enabling it to learn Latin-specific language representations that generic multilingual BERT models might not capture effectively, which offers many improvements for Latin-specific NLP tasks. Bamman and Burns [2] demonstrate the usefulness of Latin BERT with four case studies of different NLP tasks, showing Latin BERT outperforms other models. First, they show that Latin BERT produces a new state of the art for Part of Speech (POS) tagging on three different datasets. Then they show Latin BERT's ability to be used for the task of text infilling. It also outperforms static embeddings for the task of word sense disambiguation. Finally, they demonstrate Latin BERT's potential for contextual nearest neighbours.

2.4 Intent Classification

Intent classification is an important task in Natural Language Understanding (NLU) and acts as an important component of certain dialogue systems, allowing for the identification of what goal the user wants to accomplish based on natural language input [31, 33, 42]. It is commonly used in customer service chatbots, task-oriented dialogue systems, and question-answering systems. Intent classification involves categorising user utterances according to their underlying goal. The dialogue system uses the classified intent to determine what action to take. Accurate intent classification is very important to chatbot effectiveness because misclassification leads

to inappropriate responses.

Various approaches have been applied to intent classification as listed by Hoque et al. [18] and Ran et al. [30]. Traditional intent classification approaches have used machine learning methods such as Support Vector Machine (SVM), logistic regression, and Naive Bayes. Deep learning based approaches such as Recurrent Neural Networks (RNN), Gated Recurrent Units (GRU), and Long Short-Term Memory (LSTM) have also been used for this task [25, 31, 32, 44]. More recently, the use of transformer-based models, such as BERT, for intent classification has achieved strong results, performing better than other models [4, 5, 12, 15]. For instance, Chen et al. [5] propose a joint intent classification and slot filling model based on BERT, with results showing that it outperformed attention-based RNN models and slot-gated models on benchmark datasets.

A related challenge in intent classification is out-of-scope detection, which is the ability to recognise inputs that do not belong to any of the supported intents. Larson et al. [24] introduce a dataset designed to evaluate out-of-scope detection. They show how it is important that an intent classification system is able to identify out-of-scope inputs, because in real deployments of dialogue systems users will inevitably ask questions outside the system's intended scope, and not all inputs will belong to a known intent class. Zhang et al. [43] further show that even pre-trained transformer models can find it challenging, and often mistakenly classify out-of-scope inputs that semantically resemble in-scope ones.

This project adapts intent classification to the specific domain of Latin grammar by treating grammatical concepts as intents in a dialogue system. This adaptation reframes the intent classification task as identifying which grammatical aspect the learner is asking about.

2.5 Morphological Analysis

Morphological analysis is the task of tagging a word with a set of grammatical features that describe its inflectional form, such as part-of-speech, case, number, gender, tense, mood, and voice. Neural approaches to this task have commonly been framed as an extension of POS tagging, in which each token is assigned a tag that encodes its morphological properties [37]. For morphologically rich languages, the task is challenging because the number of possible feature combinations can grow quickly, leading to very large tag sets that are often sparsely represented in the training data.

Previous works have proposed different approaches for implementing the task of morphological analysis. For example, Singh et al. [34] present a morphological analyser for Ancient and Medieval Greek, and in their approach, the morphological

analyser is designed to output a single, fine-grained label for each token, which represents a full morphological analysis by combining each distinct morphological feature into a single tag. The authors recognise that this approach results in a very high number of potential unique class labels, depending on the specific treebank used for training, which makes the task of predicting the labels quite challenging. On the other hand, Tkachenko and Sirts [37] argue that treating morphological tags as monolithic labels has many limitations, and instead propose viewing them as composite labels made up of multiple morphological category values. Additionally, Inoue et al. [21] propose a multi-task approach where each morphological feature is predicted jointly, as they state that utilizing shared information from other morphosyntactic categories is beneficial. Tkachenko and Sirts [37] further build onto this idea, by proposing a hierarchical model, where the POS is predicted first and the values of all other morphological categories are then predicted as dependent on the POS tag. Additionally, Søgaard and Goldberg [35] argue that in a multitask setting, taking a hierarchical approach, where lower level tasks (such as POS tagging) feed into higher level tasks, improves model performance, because it allows for higher level tasks to benefit from the representations developed in the earlier stages.

As for morphological analysis for Latin specifically, a few previous works have applied modern neural methods to this task. The Latin BERT model of Bamman and Burns [2] discussed in Section 2.3 was shown to produce state-of-the-art results on POS tagging across three Latin treebanks. Nehrdich and Hellwig [26] extended this line of work by showing that the Latin BERT embeddings also produce competitive results on morpho-syntactic tagging. Additionally, Wróbel and Nowak [40] present a system capable of POS and morphological tagging of Latin, using a fine-tuned XLM-RoBERTa model, demonstrating that large multilingual transformer models also transfer well to Latin. Finally, Keersmaekers and Mercelis [22] systematically compare different ways of adapting transformer models to morphological tagging for Ancient Greek and Latin, including a single model with a combined monolithic feature tag, multiple separate models per feature, and a multi-task model that predicts all features using a shared encoder. They report that, while simply fine-tuning a transformer to predict a monolithic tag already achieves decent results, each of the adaptations they consider further improves tagging accuracy. However, they also show that the multi-task model did not outperform the multiple models approach, and, in some configurations, performed slightly worse, but they were unable to explain why this was the case.

3 Methodology

This chapter provides a detailed explanation of the implementation of the proposed solution to achieve the project's objectives, explaining the design decisions and their justifications by drawing from the research carried out and discussed in the previous chapter. The full implementation is available at the following GitHub repository: `https://github.com/spiteriamy/final-year-project`

3.1 Problem Formulation and System Overview

The system takes as input a natural-language English question about a single Latin word (e.g. 'What case is puellae?'), and returns an English response identifying the grammatical feature being asked about. The task is split into four stages. Given a query, the system first classifies it into an intent, extracts the inflected Latin word form from the query, retrieves the morphological information of the word, and generates a response from the intent, word, and morphological information.

The overall architecture of the system consists of several components that work and interact together to implement these stages. The main components include an intent classification module, a morphological analysis module, a response generation module, and a conversational interface. The user interacts with the system through the conversational interface by submitting a natural-language grammar question in English. The chatbot takes input in English as the students' proficiency in Latin may still be too limited to ask questions in Latin. The user's input is processed by an intent classification model that identifies the grammatical category being asked about in the question. The Latin word is extracted from the question and the system retrieves the relevant grammatical information from the morphological analysis model or the Latin words dataset. A response is generated based on the identified intent and the retrieved grammatical information by using pre-defined response templates. Finally, the chatbot returns the response to the user. An overview of the system architecture is provided by Figure 3.1.

3.2 Data and Resources

3.2.1 Latin Words Dataset

A dataset of Latin words and their grammatical features was created by using the UniMorph dataset as the base dataset and using the Whitaker's Words program to fill in and add additional grammar information to the entries. The UniMorph dataset was

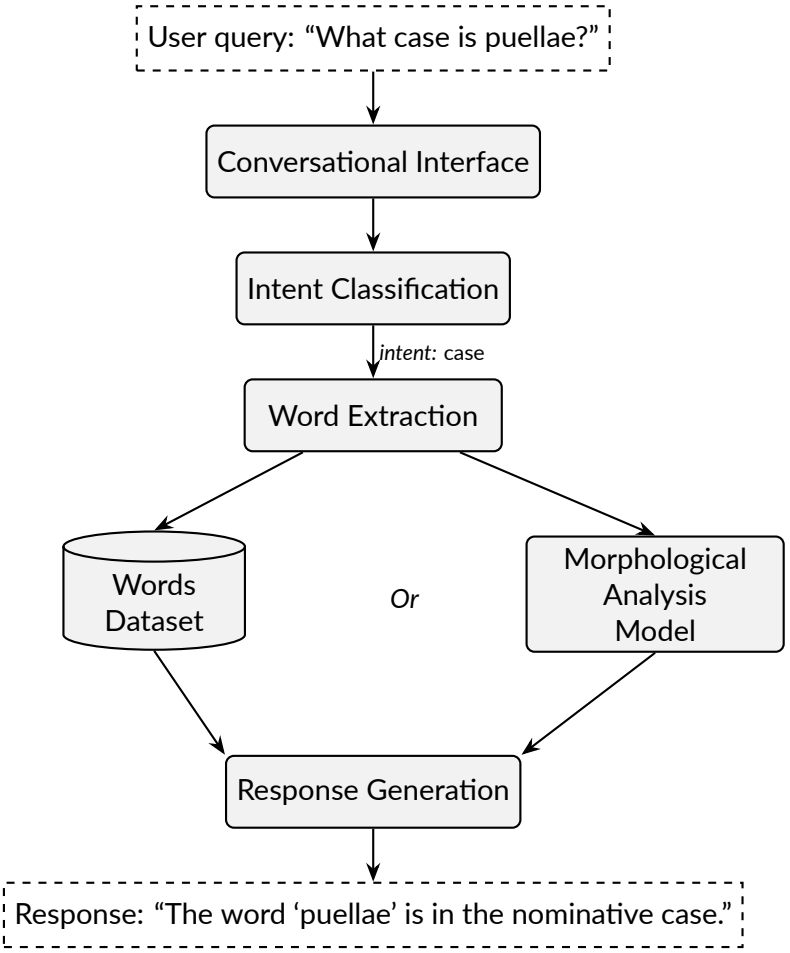


Figure 3.1 System architecture.

selected as it contains 976,489 entries of individual Latin words, along with their lemma and the grammatical features for how the word was inflected. Having a large collection of words and their corresponding grammatical features is needed to construct the intent classification dataset described in Section 3.2.2, which will contain questions about the grammar categories of specific words. While UniMorph is a good starting point for data, it has some limitations. The first limitation is that it does not include any information about the declension numbers of Latin nouns, nor does it include the conjugation number of Latin verbs. These features are important for Latin learners as they reveal the paradigms of word inflections according to different grammatical properties. Another limitation of UniMorph is that for certain parts of speech, some grammatical properties are left out. For example, gender is not included with nouns, even though Latin nouns do have a grammatical gender.

To address these limitations, Whitaker’s Words was used to fill in the gaps. This program was chosen since the output that it gives is already similar to the format of the UniMorph entries, and it includes the additional information that is missing. Since it is a command line program, it is easily interacted with and controlled programmatically, which enabled the extraction of the needed information to be automated. The result is

a CSV file of many entries of inflected Latin word forms along with their grammatical features.

Feature Extraction and Schema

The UniMorph entries store all grammatical features in a single string (e.g. `N;NOM;SG`). To enable per-feature querying, the feature string was parsed into individual columns using regular expressions: `part_of_speech`, `case`, `number`, `gender`, `tense`, `aspect`, `voice`, `mood`, `person`, `finiteness`, and `lang_specific`. The `lang_specific` column was used to store the value for entries that contained the `LGSPEC1` feature, which is for Language-Specific Features. In the Latin portion of UniMorph, this feature denotes "Supine", an etymologically rare non-finite verb form found in languages like Latin.

Supplementation with Whitaker's Words

Whitaker's Words was queried for each unique word form. For each form, the program typically returns multiple different analyses corresponding to different lemmas or inflections. The correct analysis for a given UniMorph row was selected by matching on POS and the other morphological features. This is done so that only the analysis consistent with the UniMorph entry is used to fill in missing features. Once a matching analysis was found, missing UniMorph feature values were filled (without overwriting existing ones) and the additional features (`conjugation`, `declension`, `degree`, `participle_type`, `comment`) were added as new columns.

3.2.2 Intent Classification Dataset

A second dataset, consisting of grammar-related questions and their corresponding intents, was constructed to serve as the training data for the intent classification model described in Section 3.3. The dataset was constructed from scratch, and a synthetic, template-based generation strategy was used. This strategy was chosen as collecting enough samples from the real world to produce a volume of training data sufficient for fine-tuning the model would have been too difficult and time-consuming.

Intent Categories

The intent categories defined are: `part_of_speech`, `person`, `mood`, `tense`, `voice`, `aspect`, `number`, `case`, `gender`, `declension`, and `conjugation`. Each intent corresponds to a single grammatical feature about which a learner might ask. The `person` intent is treated as a joint intent covering both person and number, since these features tend to

be asked about together for verbs. A further category, `oos` (out-of-scope), was added to allow the classifier to recognise inputs that fall outside the system's domain.

Template Creation and Question Generation

The question templates were stored in a JSON file with two top-level keys: `placeholders` and `intents`. The placeholders define the variable slots that appear in the templates, and the intents group together the templates that belong to each grammatical category. Each template consists of a question string with placeholders that are replaced with a value at generation time. The `{WORD}` placeholder is replaced with an actual Latin word form from the dataset described in Section 3.2.1. Grammatical category placeholders such as `{TENSE}` or `{CASE}` are replaced with one of a fixed list of human-readable values for the corresponding grammatical category. This is done to introduce variation without requiring a separate template per category value.

Templates were created through a combination of manual writing and with assistance from LLMs. LLMs were used to generate a variety of different ways a learner might phrase a question. They were also used to suggest paraphrased variants of the templates. This produced 341 templates across the 11 grammatical intents. Once the templates were created, the dataset itself was generated programmatically. The generation procedure is summarised below. For each intent, the generation procedure performs the following steps:

1. **Word filtering:** The Latin words dataset is restricted to rows in which all of the intent's target labels are populated. This step is important because a question such as "What tense is {WORD}?" is only meaningful, and only has a retrievable answer, if the chosen word actually has a tense. Words with missing values for the relevant feature would result in questions that could not be answered.

2. **Word sampling:** A fixed number n of words is sampled at random from the filtered subset. A fixed random seed ($\texttt{random_state} = 42$) is used throughout to ensure reproducibility. For each intent, $n = 100$ words were sampled, creating a balanced dataset across intents.

3. **Template filling:** For each sampled word, a template is selected at random from the intent's template list. The `{WORD}` slot is filled with the sampled word, and any remaining slots are filled by sampling at random from the corresponding placeholder value list.

4. **Record creation:** For each generated question, a record is written containing the intent label, the question string, the Latin word used, the values of the target labels, and the full feature string of the word.

The class balance produced by this procedure is approximate and not exact. For intents whose target labels apply to a narrow class of words, the filtered subset may contain fewer than n words, in which case the entire subset is used. However, this only occurs when generating a very large amount of questions per intent. At 100 samples per intent, the dataset produced is balanced with 100 questions per intent.

The 11 grammatical intents described above only cover inputs that the system is designed to handle. To allow the classifier to recognise other inputs, such as ambiguous questions or unsupported queries, an additional class labelled `oos` (out-of-scope) was added to the dataset. The out-of-scope examples were sampled from the CLINC150 dataset [24], an intent classification dataset consisting of 150 intents that do not overlap with the Latin grammar domain. This provided a source of realistic English utterances that are out-of-scope for the system. 70 sentences from this dataset were sampled and added to the dataset to create the `oos` class. This will ensure that this type of input is handled and the chatbot will still be able to return an appropriate response to the user, indicating that the question was not understood. The final dataset consists of 1170 total sentences.

Train, Validation, and Test Split Strategy

The synthetic generation procedure described previously produces multiple instances of each template by varying the Latin word and the values of any grammatical category placeholder. All of the generated sentences were split uniformly at random into training, validation, and test sets. 70% of the dataset was used for training, 15% for validation, and 15% for testing, resulting in 819 train, 176 validation, and 175 test sentences.

Another splitting strategy was considered for comparison, which consisted of splitting the data at the level of templates rather than sentences, holding out a subset of templates as validation-only and another as test-only, so that no sentence in the validation or test set shared its template with any training sentence. Since the first strategy described above split the dataset at a sentence-level, multiple instances of the same template could appear across all three splits, so the second strategy was an attempt to measure generalisation to unseen phrasings. However, it produced very similar performance metric results to the first strategy. The strategy used in the final system was therefore the first, to expose the model to as many phrasings as possible during training.

3.2.3 Morphological Analysis Dataset

The PROIEL Treebank was chosen as the training corpus for the morphological analyser. The treebank consists of 11,851 sentences and 110,774 word tokens. The tokens are annotated with a POS tag and a morphology tag. The morphology tag is a 10-character string, where each character denotes a value for person, number, tense, mood, voice, gender, case, degree, strength, and inflection. The schema's structure makes it straightforward to decode into named features during dataset preprocessing.

The main limitation of PROIEL for the purposes of this project is that its annotation schema does not cover all of the morphological features defined in the intent classification dataset. The three intents that are affected are `declension`, `conjugation`, and `aspect`. The 'aspect' feature is not represented as a separate feature in PROIEL like it is in UniMorph, but it is combined into the tense label. This reflects the different annotation conventions found on Latin morphology. As a consequence, the morphological analyser cannot predict these features. Additionally, the dataset is very imbalanced across classes for all features, as can be seen in Figures D.4 and D.5, which has some implications on the results as discussed in Chapter 4.

3.3 Intent Classification Model

An intent classification model was created for the system to be able to understand the user queries and identify what specific grammar category they are asking about. It is implemented as a fine-tuned BERT model, using the dataset described in Section 3.2.2, and performs single-label sequence classification over the 12 intents defined. A BERT model was selected on the basis of evidence from the literature review, which shows that it is a popular approach that produced some of the best results in comparison to other methods [18, 24]. The model was implemented using the HuggingFace `Transformers` [39] library.

The partitioning of the dataset into training, validation, and test sets is described in Section 3.2.2. Questions were tokenised with the standard BERT tokeniser, and truncated and padded to a maximum length of 50 tokens, which is comfortably above the longest template-generated question.

The main training challenge for this task was the very high performance metric results that do not reflect the true model capabilities. This is discussed further in Section 4.2. The training configuration was chosen to help the model make less overconfident predictions. Three regularisation techniques were used in combination to prevent overfitting. Hidden and attention dropout were both raised from BERT's default of 0.1 to 0.2, to encourage the model to learn more distributed features. Weight decay was set to 0.1, instead of the default 0, for the same reason. Label smoothing

with a factor of 0.1 was applied to discourage the model from becoming overconfident. The full hyperparameter values used for training are listed in Appendix A.

3.4 Morphological Analysis Model

A morphological analysis model was trained to identify the morphological structure of a given Latin word.

3.4.1 Label Representation and Preprocessing

The analyser is trained on the PROIEL Treebank, in which each token's morphology is encoded as a 10-character string. The first eight positions correspond to person, number, tense, mood, voice, gender, case, and degree. The final two positions, which correspond to strength and inflection, were discarded because none of the Latin PROIEL records carry strength information, and any tokens marked as `non-inflecting` are tokens for which no morphological prediction is required. Each morphology string was decoded into a dictionary of named feature values using a fixed lookup table per position, with absent or inapplicable positions mapped to a `None` label that is included in every feature's label vocabulary. Combined with the POS tag, this creates nine classification targets per token.

A POS to feature applicability matrix $M \in \{0,1\}^{|P|\times K}$ encodes which of the $K = 8$ morphological features are defined for each of the $|P|$ POS tags. For example, 'case' is applicable to nouns, adjectives, and pronouns but not to verbs, while 'tense' applies only to verbs. The truth values for applicability were determined from the annotations in the dataset and from the Latin language reference by Clackson [9]. This matrix is used to mask both the training loss and inference outputs, ensuring that tokens for which a given feature is undefined contribute no gradient to that feature head, and the model is not allowed to output them at inference time. The full matrix can be found in Appendix C.

For input to the encoder, sentences are tokenised with the Latin BERT tokeniser. Because PROIEL labels are word-level but the encoder operates on subwords, labels were aligned by assigning the label to only the first subword of each word and assigning the subsequent subwords an "ignore" label so they are ignored by the loss function. Sequences were padded or truncated to 128 tokens.

3.4.2 Architecture

The model consists of a Latin BERT [2] encoder followed by one POS classification head and eight POS-conditioned morphological feature heads, as illustrated in

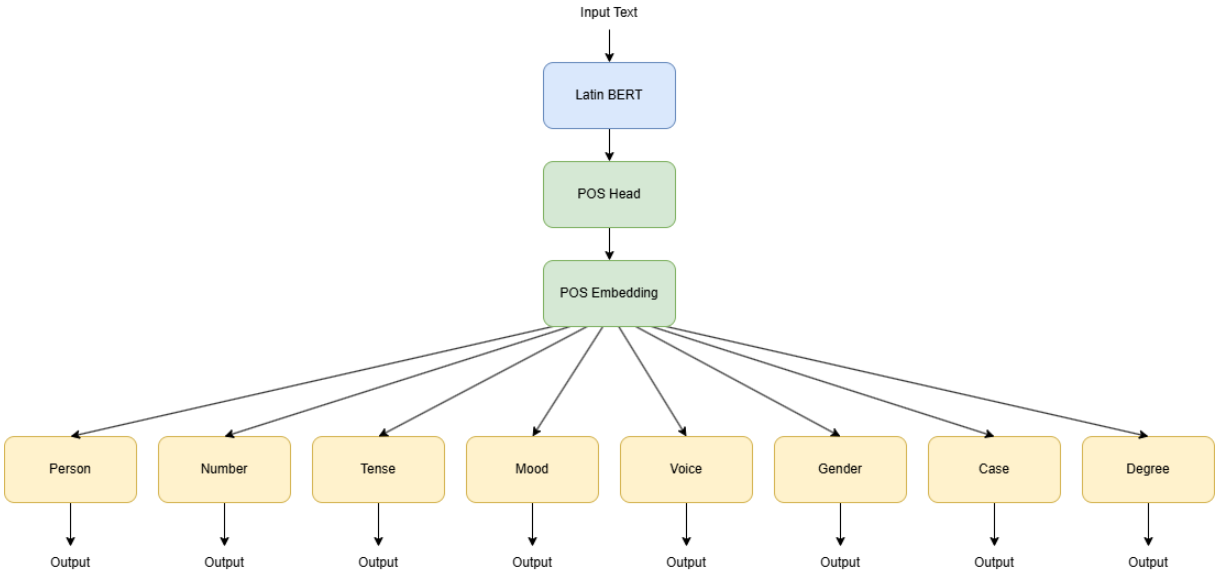


Figure 3.2 Architecture of the morphological analyser.

Figure 3.2. The model architecture consists of three stages. First, the token's contextual representation produced by the encoder is passed through a linear POS head. Second, a soft POS conditioning mechanism is used to allow the feature heads to condition their predictions on the POS decision. Third, each morphological feature head is a two-layer feed-forward network (Linear $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear) that produces logits over its label vocabulary.

3.4.3 Training

The model was trained in two phases, inspired by the approach proposed by Kumar et al. [23]. In Phase 1 the encoder was frozen and only the newly initialised components were trained. They were trained at a learning rate of 1×10^{-3} for two epochs. In Phase 2 the encoder was unfrozen and the full model was fine-tuned with a learning rate of 2×10^{-5} for up to 50 epochs with early stopping on validation POS accuracy, with a patience of 2. The hyperparameter values used for training are listed in Appendix B.

For comparison, a single stage setup was also tried, in which the entire model, including both the encoder and newly initialised heads, was fine-tuned end-to-end. However, the two-phase setup resulted in a greater improvement in fewer epochs, and was therefore used instead.

3.4.4 Confidence-Based Fallback

The analyser outputs a confidence score alongside each prediction. After training, the confidence scores per feature were examined on the validation set to assess whether

the model's confidence was predictive of its correctness, following the observation that correctly classified examples tend to receive higher prediction probabilities than misclassified ones [17]. This was done to determine a threshold, where the accuracy achieved on the subset of predictions whose confidence exceeded that threshold is at a certain target accuracy. The chosen thresholds were then applied to the held-out test set for an evaluation of their effect on performance. This evaluation is discussed in detail in Section 4.3.2.

This threshold is then used in the final system during inference so that a prediction is only accepted if its confidence exceeds the threshold, inspired by the selective classification approach discussed by Geifman and El-Yaniv [13]. Predictions that fall below the threshold are flagged as low-confidence, for fallback during the response generation. A target accuracy of 98% was selected because in a tool aimed at learners, a confident but wrong morphological tag would be more harmful. This strategy is done in two steps. First, if the POS confidence is below its own threshold, all morphological feature predictions for the token are also rejected, regardless of their individual confidences. This is done because the feature heads are conditioned on the POS, and the correctness of their predictions depends on the POS prediction being correct. Then, the confidence for each predicted morphological feature is also checked against the corresponding threshold for that feature.

3.5 Response Generation

The response generation component ties together the intent classifier and the morphological analyser to produce the system's natural-language response. The approach chosen for the response generation system is a template-based approach. Answer templates were designed for each grammar intent category. Each template is a string that contains placeholders for the insertion of the correct information. This template and retrieval approach was chosen over generating responses by using an LLM because it guarantees that any factual content in the response, which is the morphological feature value, comes directly from the analyser or the dataset to try to ensure better accuracy in the response returned. Given a user query, it identifies the Latin word being asked about, retrieves the relevant morphological feature predicted by the analyser, and inserts both into a response template selected by the classifier's predicted intent. The steps for producing a response are as follows:

1. System identifies intent.
2. System extracts the Latin word from the sentence.
3. System retrieves the morphological information for that word.

4. System constructs a template response.

3.5.1 Latin Word Extraction

User queries are English sentences containing one Latin word, for example “What case is *puellae*?”. The Latin word is extracted from the sentence by tokenising the query with the regular expression `[A-Za-zÀ-ÿ']+`, which captures alphabetic characters in the Latin script including common accented characters, and then selecting the token that is not a member of an English vocabulary. The vocabulary is taken from the Natural Language Toolkit (NLTK) `words` corpus. If exactly one non-English token is found, it is returned as the Latin word; if zero or multiple are found, the system falls back to a clarification response. This approach was chosen because both Latin and English words use the Latin alphabet, so they cannot be distinguished by their characters. An alternative approach considered was to look up words against the Latin words dataset, but this would have rejected any Latin word not present in the dataset and removed the possibility of the morphological analyser generalising to unseen word forms, which is one of its main contributions. The limitations of the chosen approach are that a few Latin words happen to also be English words, and a number of English words are also missing from the NLTK vocabulary.

3.5.2 Fallback and Error Handling

The system will fall back to a clarification message rather than output an unreliable answer in certain cases. First, if the intent classifier predicts the out-of-scope class, the system responds with a message explaining that it does not understand the input. If no Latin word is found in the sentence, the system tells the user that it could not find a Latin word and to rephrase. If the predicted intent corresponds to a feature that the morphological analyser does not support, the system retrieves the value from the Latin words dataset instead. If the word is not present in the dataset, the system tells the user that it could not find the requested information. If the morphological analyser’s confidence in the requested feature is below the per-feature threshold, it will also retrieve the value from the Latin words dataset. However, if this value is also not found in the dataset, the system will tell the user that it is not confident about its answer. Additionally, if the requested feature is inapplicable to the predicted part of speech (e.g. *tense* on a noun), the system says so to the user. This handling for these specific situations is done in order not to mislead learners with incorrect grammatical claims.

3.6 Conversational Interface

A conversational user interface (UI) was designed and implemented to allow users to interact with the chatbot by asking questions and receiving answers. This is the user-facing component of the system, which enables real-time interaction and fulfils one of the project's objectives of creating a conversational tool.

The UI is web-based, and it was designed to be user-friendly and intuitive to enhance usability. It was developed using Flask $3.1.0$, and consists of two main pages. The first is a landing page that provides users with background information about the project and instructions on how to use the chatbot. The second page is the main interface, and it contains a chat area as well as a feedback questionnaire form. The chat area allows users to submit their questions and receive responses from the chatbot. To the side of the chat area is a feedback questionnaire form where users provide feedback on the chatbot's performance and their overall experience. This feedback was used for the qualitative evaluation of the system, as discussed in Chapter 4. The UI can be seen in Figure 3.3.

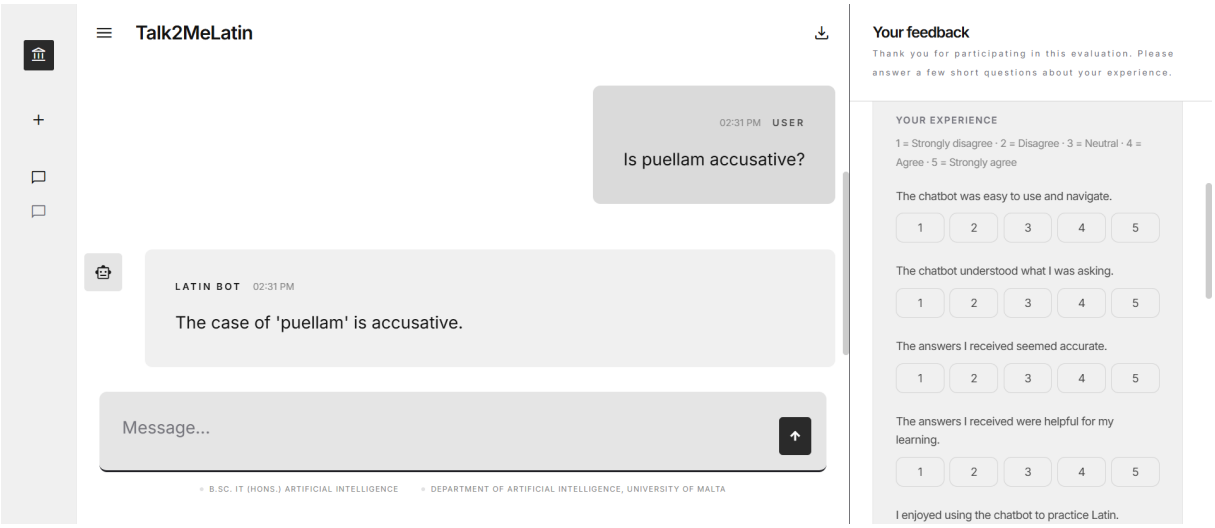


Figure 3.3 User interface.

4 Evaluation

This chapter reports the evaluation of the system described in Chapter 3. The aim of the evaluation is to provide evidence against the objectives set out in Chapter 1, both in terms of the technical performance of the system's components and in terms of its usefulness to actual Latin learners. The chapter is organised by system component, since the two main components implemented (the intent classifier and the morphological analyser) are independently trained and so must be evaluated separately, and ends with a user evaluation of the entire system.

4.1 Evaluation Setup

The evaluation is structured into two main stages. The first is a component-level technical evaluation, in which the intent classifier and the morphological analyser are assessed against held-out test data using standard classification metrics. The second is a user evaluation, in which Latin learners interact with the system and complete a questionnaire. Both parts of the evaluation together are important to assess the entire system's overall behaviour.

4.2 Intent Classification

4.2.1 Setup

The intent classifier was evaluated on the held-out test set described in Section 3.2.2, which consists of 15% of the full dataset (175 sentences). Performance was measured using standard classification metrics, these being accuracy, precision, recall, and F1 score, reported both per class and as averages across classes. Additionally, a confusion matrix is also reported. Precision is used to see the correctness of the relevant items that were identified by the model while recall is used to see how effective the model is at identifying the relevant items. The F1 score is the harmonic mean of precision and recall and provides a balance of both.

4.2.2 Results and Discussion

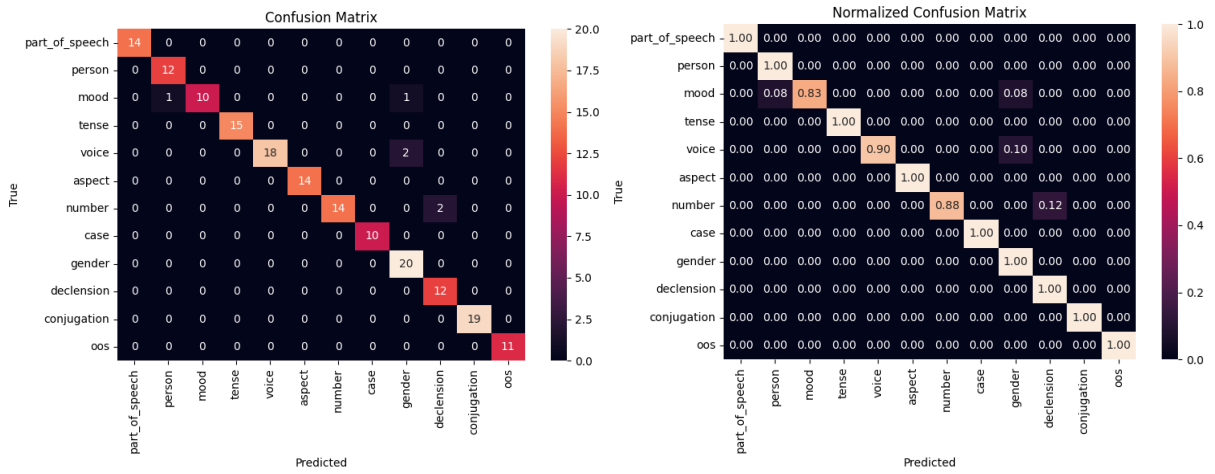
Table 4.1 presents the per-class and overall accuracy, precision, recall, and F1 results of the intent classification model on the test set, and Figure 4.1 shows the corresponding confusion matrices. From these results it can be seen that the classifier reaches both an overall accuracy and F1 score of 0.97. Six of the twelve intents achieve perfect

Intent	Precision	Recall	F1
part_of_speech	1.00	1.00	1.00
person	0.92	1.00	0.96
mood	1.00	0.83	0.91
tense	1.00	1.00	1.00
voice	1.00	0.90	0.95
aspect	1.00	1.00	1.00
number	1.00	0.88	0.93
case	1.00	1.00	1.00
gender	0.87	1.00	0.93
declension	0.86	1.00	0.92
conjugation	1.00	1.00	1.00
oos	1.00	1.00	1.00
Macro Average	0.97	0.97	0.97
Weighted Average	0.97	0.97	0.97
Overall Accuracy	0.97		

Table 4.1 Intent classification metric results.

scores, and the remaining six show varying F1 scores between 0.91 and 0.96, reflecting a small number of confusions seen in Figure 4.1. The overall results show that the classifier can handle the twelve intents well on data drawn from the templates.

However, these results should be interpreted with the following limitation kept in mind. The training data is generated only from a finite set of templates. Therefore, the reported metrics may overstate the classifier's ability to handle phrasings unlike those that it was trained on. The test set is obtained from the same templates that generated the training data, so the results show the classifier's ability to recognise the templated phrasings, rather than its ability to handle the open-ended natural language a real learner would use. Therefore, the user evaluation provides more direct evidence on the performance of the model and its ability to handle actual learner questions. This is further discussed in Section 4.4, where it can be seen that the lowest-scoring item was *"the chatbot understood what I was asking"*, with 0% agreement. The contrast between a 0.97 test set accuracy and a 0% user evaluation agreement on understanding shows that there is a gap between the questions that the model can identify and the variety of questions that users may ask. Addressing this issue would be crucial in order to improve performance in understanding real user questions.



(a) Confusion Matrix

(b) Normalised Confusion Matrix

Figure 4.1 Intent classification confusion matrices.

4.3 Morphological Analyser

4.3.1 Setup

The morphological analyser was evaluated on the held-out test split of the PROIEL Treebank (16,423 tokens). Per-feature performance was measured using accuracy, macro F1, and weighted F1 for each of the nine features: POS, person, number, tense, mood, voice, gender, case, and degree. Additionally, full-tag accuracy is also reported, which is the proportion of tokens for which all predicted features are simultaneously correct. This is reported as a stricter measure and would typically be lower than a single per-feature accuracy. Overall, the data used is quite imbalanced across classes for all of the features, as can be seen in Figures D.4 and D.5, which is why precision, recall, and F1 are reported alongside accuracy. Both weighted and macro averages are compared to evaluate the impact that the class imbalance has on the model. The macro average treats each class equally, so poor performance on minority classes will affect the score. The weighted average assigns each class a weight proportional to its number of instances, and can hide poor performance on rare classes. Having a high weighted average score while the macro average is low would indicate that the model is performing well on the majority class but failing on the rare classes.

4.3.2 Results and Discussion

Morphological Analysis

Table 4.2 reports per-feature performance on the PROIEL test set, together with the full-tag accuracy. The support column gives the number of test tokens for which the

Feature	Support	Accuracy	Macro Precision	Weighted Precision	Macro Recall	Weighted Recall	Macro F1	Weighted F1
POS	16,423	0.76	0.76	0.78	0.56	0.76	0.62	0.75
Person	5,541	0.79	0.77	0.79	0.73	0.79	0.75	0.79
Number	11,542	0.79	0.81	0.79	0.66	0.79	0.71	0.78
Tense	4,073	0.71	0.67	0.70	0.54	0.71	0.58	0.70
Mood	4,073	0.80	0.49	0.80	0.45	0.80	0.47	0.80
Voice	4,073	0.90	0.84	0.90	0.67	0.90	0.73	0.89
Gender	7,469	0.71	0.46	0.69	0.31	0.71	0.34	0.69
Case	7,469	0.74	0.78	0.76	0.64	0.74	0.69	0.74
Degree	2,143	0.88	0.65	0.88	0.54	0.88	0.57	0.88
Full-tag	16,423	0.56	–					

Table 4.2 Per-feature and full-tag performance of the morphological analyser on the test set.

feature applies (features such as *tense* and *mood* only apply to verbs, while *case* only applies to nominals). Full per-class results for all features can be seen in Appendix D. Several issues can be noted from these results. First, the model achieved a full-tag accuracy score of 0.56, meaning only around half of the test tokens received a correct full morphological analysis of all applicable features. This can probably be attributed to the varying performance of each individual feature, with some features achieving very low scores, meaning they are each individually less reliably predicted, contributing to the overall low full analysis accuracy. For each individual feature, the model achieved different accuracies ranging from 0.71 to 0.90. However, as mentioned previously, the data used is imbalanced so looking at just the accuracy is misleading of the actual performance, which is better interpreted by looking at the other results. Table 4.2 shows that there is a large gap between macro and weighted F1 for many features, most notably *mood*, *degree*, and *gender*. These results show that the model is performing poorly on the minority classes, and it is achieving the higher weighted F1 and accuracy scores by just predicting the dominant class. This is an issue since in a system aimed at learners the model should be able to identify even the rarer forms, therefore this would need to be addressed in order to ensure the correctness of the responses.

Confidence-Based Fallback

This part of the evaluation describes the analysis that was done to determine the thresholds for the morphological analyser's confidence based fallback described in Section 3.4.4. This analysis addresses three aspects. First is whether the model is calibrated, meaning does a predicted probability of p correspond to an actual accuracy

near p [10]. Second is whether the correct and incorrect predictions are separable by their confidence values. A correct prediction should be more likely to have a higher confidence value, while an incorrect prediction should have a lower confidence value, so a threshold can be found to detect the errors [17]. Finally is what thresholds achieve a useful trade-off between accuracy on the retained predictions and the proportion of predictions retained (the coverage) [41]. To determine if the model was calibrated, reliability diagrams and the Expected Calibration Error (ECE) metric were used to see how closely the model's confidence aligns with the actual accuracy [14, 27, 29]. Additionally, correct versus incorrect confidence histograms, and accuracy-coverage curves were also produced. Thresholds were then selected per feature at three target accuracy levels, these being 90%, 95%, and 98%. For each target, the lowest confidence threshold at which the accuracy of the retained predictions reaches the target was chosen, since this is the threshold that maximises coverage at the accuracy constraint. All of this was done on the validation set, in order to perform the threshold selection itself. The thresholds chosen here are then applied and evaluated on the held-out test set, to see their corresponding effect on performance.

Feature	No.	Accuracy	ECE	Mean Conf.	Conf. when correct	Conf. when wrong	Separation
POS	16,706	0.7551	0.0339	0.7886	0.8758	0.5197	0.3561
Person	5,500	0.7784	0.0752	0.8519	0.9104	0.6464	0.2639
Number	11,720	0.7916	0.0334	0.8211	0.8602	0.6729	0.1873
Tense	4,037	0.6973	0.0712	0.7669	0.8332	0.6143	0.2189
Mood	4,037	0.7917	0.0737	0.8639	0.9087	0.6938	0.2149
Voice	4,037	0.8960	0.0272	0.9098	0.9208	0.8156	0.1052
Gender	7,683	0.7092	0.0309	0.7382	0.8077	0.5686	0.2391
Case	7,683	0.7405	0.0641	0.8035	0.8694	0.6153	0.2541
Degree	2,215	0.8763	0.0384	0.9102	0.9369	0.7209	0.2161

Table 4.3 Per-feature confidence overview on the validation set.

Table 4.3 reports per feature the number of predictions evaluated, overall accuracy, ECE, the mean confidence, and the mean confidence when the prediction is correct versus incorrect. The 'Separation' column gives the difference between the mean confidence on correct and incorrect predictions. This number is important because a feature head with a small separation is similarly confident both when right and when wrong, so larger values indicate that confidence is more informative about correctness. The ECE is computed with 15 equal-width confidence bins. Figure D.1 shows reliability diagrams for each feature head. A perfectly calibrated head would result in a diagonal line, and points below the diagonal indicate overconfidence, meaning it is right less often than it claims, while points above indicate underconfidence [29]. Additionally, Figure D.2 overlays the confidence distributions of

Feature	Target 90%		Target 95%		Target 98%	
	t	Coverage	t	Coverage	t	Coverage
POS	0.597	72.01%	0.773	62.21%	0.914	53.42%
Person	0.743	72.93%	0.919	60.80%	0.989	49.33%
Number	0.733	64.78%	0.873	50.09%	0.964	39.61%
Tense	0.803	49.54%	0.914	38.92%	0.969	32.75%
Mood	0.793	71.32%	0.944	56.75%	0.994	38.57%
Voice	0.552	98.61%	0.909	63.71%	0.974	40.05%
Gender	0.703	53.30%	0.863	40.47%	0.964	28.96%
Case	0.763	62.29%	0.914	50.24%	0.984	37.72%
Degree	0.567	94.76%	0.838	77.79%	0.959	65.46%

Table 4.4 Per-feature confidence thresholds t that first achieve each target accuracy on the validation set, together with their coverage (proportion of predictions retained).

correct and incorrect predictions per feature. A lot of overlap at high confidence would mean that the model is confidently wrong on a large amount of inputs, and the threshold would only reduce errors with a large coverage loss. Overall, the per-feature ECE and reliability diagram results indicate that no feature is severely miscalibrated, as the ECE values range from 0.0272 to 0.0752, and most curves stay around the diagonal. However, the reliability curves do also show that none of the features are perfectly calibrated, with many curves falling below the diagonal indicating that there is some overconfidence. Some features are clearly more overconfident than others, for example *case*, *person*, *mood*, and *tense* have a relatively high ECE and calibration curves that are lower below the diagonal. Looking at the difference in confidence between the correct and incorrect predictions, it can be seen that these values vary across features. POS has the highest separation, so its confidence on predictions is the most informative in separating correct from incorrect predictions. This is good since the system checks for the prediction's confidence against the POS threshold first when deciding when to accept or reject the prediction, and so a reliable threshold for the POS feature is important. On the other hand, the *voice* feature is highly confident both when right and when wrong. This can also be seen clearly in Figure D.2, where there is a lot of overlap at high confidence when compared to other features.

Additionally, Figure D.3 shows the accuracy of the retained predictions against the coverage at each confidence threshold from 0 to 1 for each feature. The dashed vertical lines mark the thresholds that first achieve 90%, 95%, and 98% accuracy. Table 4.4 reports the resulting thresholds together with their coverage at each target. It can be seen that the coverage varies greatly between all features. At the highest threshold, gender retains the lowest amount of predictions which aligns with the

results shown in Table 4.2 above, as gender is the worst performing feature overall. Meanwhile, degree and POS retain the most, which can be explained by Figure D.2, where for degree it can be seen that the correct predictions are concentrated strongly at confidence near 1.0, so the wrong predictions are removed by the threshold without taking many correct predictions. For POS the correct and wrong predictions are highly separated, with most of the correct predictions clustered at a high confidence and most wrong predictions near the lower end.

Feature	Support	Accuracy	Macro Precision	Weighted Precision	Macro Recall	Weighted Recall	Macro F1	Weighted F1
POS	8827 (53.75%)	0.98	0.90	0.98	0.84	0.98	0.86	0.98
Person	2773 (50.05%)	0.99	0.99	0.99	0.98	0.99	0.98	0.99
Number	4700 (40.72%)	0.98	0.98	0.98	0.96	0.98	0.97	0.98
Tense	1327 (32.58%)	0.98	0.57	0.98	0.52	0.98	0.54	0.98
Mood	1559 (38.28%)	0.98	0.55	0.98	0.53	0.98	0.54	0.98
Voice	1692 (41.54%)	0.98	0.65	0.98	0.62	0.98	0.63	0.98
Gender	2259 (30.25%)	0.98	0.49	0.98	0.49	0.98	0.49	0.98
Case	2954 (39.55%)	0.98	0.99	0.98	0.90	0.98	0.93	0.98
Degree	1440 (67.20%)	0.98	0.74	0.98	0.68	0.98	0.71	0.98
Full-tag	6365 (38.76%)	0.98	—					

Table 4.5 Performance of the morphological analyser on the test set using per-feature thresholds.

Finally, the effect of using the feature thresholds on performance was evaluated on the test set. The highest thresholds for a target accuracy of 98% were used, as these were the ones selected for the system as explained in Section 4.3.2. Table 4.5 reports the performance per feature when using the thresholds, to see what per-feature thresholding alone contributes, while Table 4.6 reports the performance when using the two step threshold comparison described in Section 4.3.2, where a token's features are rejected entirely if POS confidence is below its threshold, then each feature is checked against its own threshold. The support column gives the number of test tokens for that feature that were retained after applying the thresholds, along with the percentage remaining out of all test tokens for that feature. The metrics were computed on the subset of retained tokens only. The full-tag accuracy is

Feature	Support	Accuracy	Macro Precision	Weighted Precision	Macro Recall	Weighted Recall	Macro F1	Weighted F1
POS	8827 (53.75%)	0.98	0.90	0.98	0.84	0.98	0.86	0.98
Person	2449 (44.20%)	0.99	0.99	0.99	0.98	0.99	0.99	0.99
Number	3814 (33.04%)	0.99	0.99	0.99	0.98	0.99	0.99	0.99
Tense	1159 (28.46%)	0.99	0.57	0.99	0.55	0.99	0.56	0.99
Mood	1365 (33.51%)	0.99	0.55	0.99	0.54	0.99	0.54	0.99
Voice	1412 (34.67%)	0.99	0.65	0.99	0.63	0.99	0.64	0.99
Gender	1890 (25.30%)	0.98	0.49	0.98	0.49	0.98	0.49	0.98
Case	2303 (30.83%)	0.99	0.99	0.99	0.92	0.99	0.94	0.99
Degree	1041 (48.58%)	1.00	0.75	1.00	0.75	1.00	0.75	1.00
Full-tag	6365 (38.76%)	0.98	—					

Table 4.6 Performance of the morphological analyser on the test set using two-step threshold comparison.

computed over fully accepted tokens, where every single feature passed. It can be observed that every feature reaches an accuracy close to the 98% target on the held-out test set under both thresholding schemes. When compared to when no thresholds are applied, it can be seen that the full-tag accuracy rises from 0.56 to 0.98 once the thresholds are applied, at a coverage of 38.76%.

Comparing the two thresholding schemes, it can be seen that per-feature thresholding on its own already brings every feature to the target, and there is not a very large difference in per-feature results between the two. However, the two-step approach then has a reduction in coverage. The very small improvement in performance with a more noticeable reduction in coverage shows that the check on POS first may not be very useful. The full-tag coverage and accuracy is identical in both, since a complete correct analysis already requires the POS prediction to be accepted.

It can also be seen from both tables 4.5 and 4.6 that the macro averaged scores remain quite low on several features, notably Gender, Mood, and Tense, even though the weighted F1 reaches 0.98. Tokens from the minority classes were not being retained after using the thresholds, because the model performs poorly on minority classes and is less confident in its predictions for them. The thresholds therefore are

filtering out the minority class instances, leading to many classes with zero support, contributing to low macro average scores. This also further highlights the model's stronger performance on majority classes and weaker performance on rare classes.

Effectiveness of POS Conditioning

To test whether the POS conditioning described in Section 3.4.2 actually improves the results or not, a variant was trained in which the POS context embedding is removed from the input to each feature head, so the feature heads receive only the Latin BERT representation. The two models share the same training schedule, data splits, and hyperparameters, and are different only in this aspect. Table 4.7 compares per-feature accuracy and F1 with and without POS conditioning.

Feature	Without POS conditioning			With POS conditioning		
	Accuracy	Macro F1	Weighted F1	Accuracy	Macro F1	Weighted F1
pos	0.7611	0.6199	0.7529	0.7612	0.6190	0.7530
person	0.7921	0.7486	0.7882	0.7944	0.7519	0.7907
number	0.7909	0.7123	0.7785	0.7899	0.7105	0.7777
tense	0.7100	0.5813	0.7003	0.7135	0.5845	0.7037
mood	0.8055	0.4639	0.8004	0.8031	0.4693	0.7976
voice	0.9033	0.7054	0.8907	0.9033	0.7340	0.8911
gender	0.7059	0.3456	0.6854	0.7054	0.3435	0.6856
case	0.7439	0.6966	0.7413	0.7427	0.6937	0.7401
degree	0.8829	0.6032	0.8777	0.8824	0.5673	0.8765

Table 4.7 Per-feature results on the test set with and without POS conditioning.

These results reveal that overall the POS conditioning did not contribute to an effect on the results. For seven of the nine features the difference in results between the conditioned and unconditioned variants are very small. As expected, POS itself is unchanged since the POS head receives no conditioning input from itself. The remaining feature heads do not show any pattern in improvement or decrease of performance, and most changes are near zero. Only two features had a slightly larger change in effect. Voice macro F1 improves by $+0.0286$ under conditioning, while degree macro F1 decreases by -0.0359. Both of these effects exceed the changes observed for any other feature, however they still do not contribute to a very significant change. Overall, this experiment does not provide strong evidence that the POS conditioning is helpful enough to earn its complexity. An explanation could be that the representations produced by Latin BERT already carry enough POS information for the feature heads to use, so adding the POS embedding provides little

information that they do not already have. These results indicate that the simpler architecture without conditioning could be an alternative for future attempts of the system, and that any future reintroduction of POS conditioning should be motivated by stronger evidence than this comparison provides.

4.4 User Evaluation

4.4.1 Setup

A user evaluation was conducted as the final component of the assessment to see whether the system is genuinely useful to actual Latin learners. A total of eight participants were recruited. These participants consisted of a combination of University of Malta students currently studying Latin, as well as members of online Latin learning communities. Participants were asked to interact with the system freely for approximately 15 minutes, asking it questions of their choice. They were also asked to complete a questionnaire to provide feedback on their experience with using the chatbot.

The questionnaire was divided into three sections. The first collected two background variables, self-reported Latin proficiency (no prior experience, beginner, intermediate, advanced, fluent/professional) and frequency of AI chatbot use (never, rarely, sometimes, often, daily), to provide more context to the following responses. The second section presented seven Likert scale questions, on a 5-point scale (Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree). These questions were for evaluating the usability, perceived correctness, and perceived usefulness of the chatbot, as well as enjoyment and motivation when using the chatbot. The questions presented in this section were adapted from the Student Attitude Questionnaire presented by Alenezi and Alenezi [1], who also set out to evaluate how effective chatbots are in assisting with language learning. The final section collected free text responses to three open-ended questions, which were what the participant liked most, what could be improved, and any further comments that the participant would like to add, in order to identify in more detail any limitations, areas of improvement and future work. The full questionnaire is provided in Appendix E. The Likert items were chosen to address four different aspects of the system's performance. The item *"The chatbot was easy to use and navigate"* assesses *Usability*. *Perceived correctness* was addressed by *"The chatbot understood what I was asking"* and *"The answers I received seemed accurate"*. Two separate items were used to be able to assess the system's input understanding separately from the correctness of its outputs. *Perceived pedagogical usefulness* was assessed by *"The answers I received were helpful for my learning"* and *"Using the chatbot*

increased my interest in learning Latin". Finally, *Endorsement* was measured by *"I would recommend this tool to other Latin learners"*.

4.4.2 Results and Discussion

Of the eight participants, the proficiency and chatbot-use distributions are summarised in Table 4.8. Overall, half of the participants identified as advanced, three as intermediate, and one as beginner. Table 4.9 summarises the responses to the seven Likert items, reporting the median, mean, standard deviation, and the proportion of responses scoring four or five (agreement). The corresponding distributions for each item can also be seen in detail in Figure 4.2. The results revealed that, overall, the participants did not find the system useful for their learning. The negative result is consistent across items and the aspects that were addressed. Of the four aspects that the questionnaire was designed to measure, only *usability* comes out positive. *Perceived correctness*, *perceived pedagogical usefulness*, and *endorsement* all fall at or below the mid-point of the scale.

Variable	**Category**	**Count**	**%**
Latin proficiency	No prior experience	0	0.0
	Beginner	1	12.5
	Intermediate	3	37.5
	Advanced	4	50.0
	Fluent / professional	0	0.0
Chatbot use	Never	0	0.0
	Rarely	3	37.5
	Sometimes	3	37.5
	Often	1	12.5
	Daily	1	12.5

Table 4.8 Background characteristics of the user evaluation sample.

The lowest scoring item overall is *"The chatbot understood what I was asking"* and it is also the most consistently rated negatively across participants. The fact that this item scored lower than *"The answers I received seemed accurate"* also shows that input understanding is worse than output accuracy. The high test set scores of the intent classifier discussed above therefore were not evidence that it generalises well to natural learner phrasings, but only that it can recognise phrasings similar to the ones in the template set. This also indicates that the templated distribution does not match or fully represent the full variety of possible real learner questions. The open-ended feedback from the participants further supported these results, with multiple

Item	Median	Mean	SD	Agree (%)
The chatbot was easy to use and navigate.	4.5	4.12	1.13	75.0%
The chatbot understood what I was asking.	2.0	2.12	0.83	0.0%
The answers I received seemed accurate.	3.0	2.50	1.07	12.5%
The answers I received were helpful for my learning.	2.0	2.38	1.19	25.0%
I enjoyed using the chatbot to practice Latin.	2.5	2.38	1.06	12.5%
Using the chatbot increased my interest in learning Latin.	2.0	2.25	1.04	12.5%
I would recommend this tool to other Latin learners.	3.0	2.50	1.07	12.5%

Table 4.9 Summary of Likert-item responses ($1 =$ strongly disagree, $5 =$ strongly agree).

participants reporting that sometimes the system did not understand their questions, and one participant mentioning that it was difficult to structure questions in a way to get the desired answer. Upon further inspection of some of the questions actually asked by participants, it could be noted that many of them were out of the defined types of questions that the chatbot could handle, for example requesting translation or explanation of a word in English, requesting for the explanation or comparison of concepts, or requests to inflect a word to express other forms. Since these capabilities fall outside the system's stated scope, these types of questions were not able to be answered, contributing to the low understanding score. From the open-ended feedback, participants also expressed that they would like additional features, with suggestions for improvement mentioning support for translating Latin words into English, as well as support for analysing multiple words or full sentences at once. Although many of the questions asked that did actually fall into one of the defined intents were identified as the right intent, when considering all the other types of questions asked and the feedback received, it is clear that the eleven defined intents were not enough to cover the variety of questions users may ask, which limited users. These results suggest that the most important improvement target for the system is improving the understanding component of the system, since until the intent classifier can recognise all the queries learners actually ask, it is difficult to assess the quality of the responses as they would apply only to a small amount of queries that actually make it past the intent classification stage. Additionally, when considering what users actually asked the chatbot and the features they suggested, it shows that the expectations users bring to a Latin chatbot extend beyond morphological analysis of single forms, and the defined intents are also quite restrictive.

The items related to *Perceived pedagogical usefulness* scored the lowest after

understanding, meaning no learning gain was directly measured. This is likely related to how well the system understood and answered the questions participants asked, because a learner whose questions were not understood did not receive an answer to those questions. Therefore, the low scores may partly be due to the low question understanding.

Regarding the perceived correctness of the morphological answers, the open-ended feedback revealed that at times the chatbot returns errors. One participant mentioned that it was unable to identify non-indicative moods, including subjunctives and infinitives. This aligns with the quantitative results reported in Table 4.2, where it can be seen that mood achieves a macro F1 score of only 0.47 compared to the weighted F1 of 0.80, indicating that the feature is failing on the minority classes. When looking further into the results per class for the mood feature shown in Table D.5, it can also be seen that the class distribution is severely imbalanced with the indicative class acting as the majority while the other classes are underrepresented, contributing to the lower F1 results for these classes. This pattern is noticeable for all of the features, where the heavy class imbalance is limiting the system from being able to identify minority classes reliably. This is particularly problematic given the system's intended pedagogical use, because the model must be capable of producing accurate analyses for any form a learner might encounter, not just the most frequent ones. Additionally, learners might be more likely to seek assistance with rarer or less familiar inflections, because these are the forms about which they are least confident. Therefore, a system that does not perform well on minority classes risks failing users where reliable morphological support is most needed.

Another issue pointed out by a participant was that when asking for what conjugation *amo* belongs to, the chatbot returned that it could not find this information. 'Conjugation' is one of the features that is looked up directly in the Latin words dataset, as it is not one of the features that is recognised by the morphological analyser. This indicates that there is an issue with the wordform annotations within the words dataset itself. Analysing this specific error further, it was noticed that the form *amo* appears twice in the dataset, both labelled as a noun. The word *amo* in Latin is commonly known as a verb ("to love"), and it is normally the first verb taught to Latin students. However, in Medieval Latin, *amo* can also refer to a noun meaning "cup" or "measure for wine and other liquids" (see the definition in the Medieval Latin dictionary, Niermeyer [28]). Therefore, the UniMorph Latin dataset seems to include Medieval forms while at the same time excluding more commonly known Classical forms. This can be an issue as Classical Latin is the standard version of the language that is most commonly taught to students, meaning that most users will be more familiar with this version and most questions will probably be referring to Classical Latin.

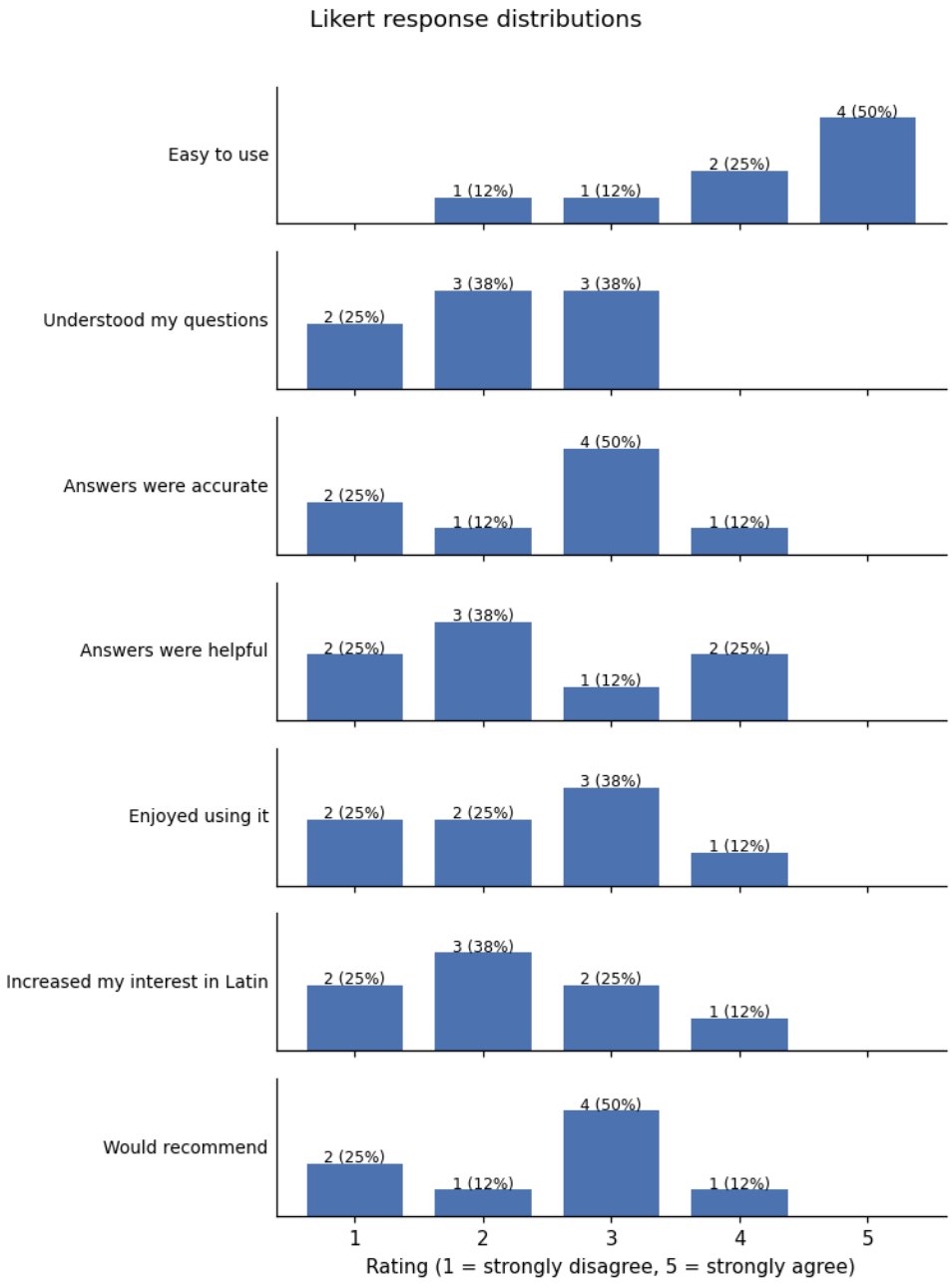


Figure 4.2 Stacked distribution of Likert responses for each questionnaire item. Each bar shows the proportion of participants selecting each response category.

5 Conclusion

5.1 Revisiting the Aims and Objectives

The main aim of the project was to develop a conversational chatbot tool for Latin learners that is able to answer questions related to Latin grammar. Overall, a prototype chatbot tool was successfully created throughout this project, combining the intent classification and morphological analysis components into a user-friendly UI, as detailed in Chapters 3 and 4. In order to have achieved this aim, the work carried out throughout this project achieved the following outcomes for the objectives stated in Section 1.3:

- **O1** - The two required datasets were successfully constructed. A Latin words dataset was built by extending the UniMorph Latin resource with the additional grammatical features needed by the system, and a synthetic intent classification dataset was generated from templates covering eleven grammatical intents plus an out-of-scope class. The intent classification dataset was used to train the intent classifier, and the Latin words dataset was used both to fill in the templates in order to generate the intent classification dataset, as well as to retrieve the required morphological information for a response when the morphological analyser provides a low confidence prediction.
- **O2** - Both models were fine-tuned as shown in Chapter 3 and evaluated in Chapter 4. The intent classifier reached an overall accuracy and F1 of 0.97 on the held-out test set. The morphological analyser managed to reach per-feature accuracies ranging between 0.71 and 0.90, with a full-tag accuracy of 0.56. A confidence-based fallback was also implemented per feature to allow the system to reject low-confidence predictions made by the morphological analyser.
- **O3** - A response generation component and a conversational UI were implemented as described in Chapter 3. The system combines the predicted intent and the morphological analysis to produce a natural-language answer to the user's question, and was successfully used by participants throughout the user evaluation reported in Section 4.4.
- **O4** - The system was evaluated in two stages, as described in Chapter 4. A component-level technical evaluation was conducted to assess the performance of the two models developed. The user evaluation collected feedback from eight Latin learners through a structured questionnaire and open-ended questions.

5.2 Critiques and Limitations

Several limitations affect the interpretation of the results and the usefulness of the system in practice. The intent classifier was trained on synthetic data produced from a small number of templates, and was evaluated on a test set produced from the same templates. The reported 0.97 accuracy therefore measures the model's ability to recognise questions similar to the templated phrasings rather than the open-ended natural language a real learner would use, which was confirmed in the user evaluation. The set of supported intents is also restrictive and the user evaluation revealed that learners commonly want additional capabilities. The morphological analyser was trained and evaluated only on the PROIEL Treebank, which is heavily imbalanced across classes for every feature. As discussed in Section 4.3.2, the model achieves higher weighted F1 and accuracy by relying on the dominant class, while its macro F1 reveals weak performance on minority classes. Additionally, the Latin words dataset constructed from UniMorph seems to also have annotation issues. As discussed in Section 4.4, the form *amo* is annotated according to its meaning in Medieval Latin, while the more commonly known Classical form is excluded. This issue was also identified only because a participant happened to ask the relevant question, so there may be other annotation issues of the same kind in the dataset that remain undetected.

5.3 Future Work

Based on the findings discussed in Chapter 4, several areas for improvement of the developed chatbot system have been identified. In order to improve the system's question understanding, the intent classifier could be trained using more diverse data than the current template generated data. Collecting a corpus of real questions from Latin learners would be beneficial for this as the resulting data would be more authentic. Additionally, question understanding can be further improved by extending the set of question types the system can handle to reflect what learners requested during the user evaluation. Additional capabilities such as translation of Latin words into English, analysis of multiple words or full sentences, inflection of a given word form into other forms, and conceptual explanations of grammatical categories all appeared in the participant feedback. Adding support for these question types would increase the chatbot's usefulness to learners. Furthermore, the performance of the morphological analyser on minority classes could also be addressed to improve results on rarer word forms. Finally, the Latin words dataset could also be revised to identify any annotation issues and prioritise Classical forms, which are what is most commonly taught to learners. This would be more ideal as this aligns more with learners'

expectations. These improvements would help in enhancing the accuracy and overall usability of the system, providing a more helpful tool for Latin language learners.

5.4 Final Remarks

In conclusion, this project has presented the development and evaluation of a prototype conversational chatbot that answers Latin grammar questions by combining an intent classifier and a morphological analyser, making a step towards the development of a conversational tool that can help Latin learners. Through the evaluation, the areas where the system can be improved have been identified, providing groundwork for future work.

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Appendix A Intent Classifier Hyperparameters and Training Configuration

This appendix lists the full hyperparameter and training configuration for the intent classification model described in Section 3.3, together with the software versions used to train and evaluate it. Default parameters from the HuggingFace `Transformers` `Trainer` were kept except where stated.

Parameter	Value
Model	
Base model	`bert-base-uncased`
Number of labels	12
Hidden dropout	0.2
Attention dropout	0.2
Tokenisation	
Tokeniser	BERT WordPiece (uncased)
Maximum sequence length	50
Padding	to maximum length
Truncation	enabled
Optimisation	
Optimiser	AdamW (default)
Learning rate	4×10^{-5}
Learning-rate scheduler	Linear with warmup
Warmup proportion	0.1
Weight decay	0.1
Label smoothing	0.1
Mixed precision	fp16
Training schedule	
Per-device batch size	32
Gradient accumulation steps	1
Maximum epochs	100
Evaluation strategy	every epoch
Checkpoint strategy	every epoch
Early-stopping patience	2 epochs
Early-stopping metric	validation loss
Best-model selection	lowest validation loss
Data splits	
Train / validation / test	70% / 15% / 15%
Random seed	42

Table A.1 Hyperparameters and training configuration for the intent classifier.

Software and Hardware Environment

Training and evaluation were carried out in a Google Colab environment. Table A.2 lists software and hardware used. Library versions correspond to the Colab default at the time of training.

Component	Version
Python	3.12.13
PyTorch	2.10.0+cu128
Transformers	5.0.0
Datasets	4.0.0
scikit-learn	1.6.1
pandas	2.2.2
NumPy	2.0.2
Matplotlib	3.10.0
seaborn	0.13.2
Compute environment	Google Colab
GPU	single NVIDIA T4
CUDA	12.8

Table A.2 Software and hardware environment used for training and evaluating the intent classifier.

Appendix B Morphological Analyser Hyperparameters and Training Configuration

This appendix lists the full hyperparameter and training configuration for the morphological analyser described in Section 3.4, together with the software versions used to train and evaluate it. The model is fine-tuned in two phases: Phase 1 trains only the prediction heads on top of a frozen Latin BERT backbone, and Phase 2 unfreezes the backbone for full fine-tuning. Default parameters from the HuggingFace `Transformers` `Trainer` were kept except where stated.

Parameter	Value
Model	
Base encoder	Latin BERT
Number of heads	9 (POS + 8 morphological features)
POS embedding dimension	64
Dropout	0.1
Tokenisation	
Tokeniser	Latin BERT WordPiece
Padding	to maximum length
Truncation	enabled
Optimisation (shared)	
Optimiser	AdamW (default)
Weight decay	0.01
Mixed precision	fp16
Training schedule	
Per-device train batch size	32
Per-device eval batch size	32
Gradient accumulation steps	1
Evaluation strategy	every epoch
Checkpoint strategy	every epoch
Best-model selection metric	POS accuracy (higher is better)
Phase 1: heads only (BERT frozen)	
Learning rate	1×10^{-3}
Maximum epochs	2
Phase 2: full fine-tune	
Learning rate	2×10^{-5}
Maximum epochs	50
Early-stopping patience	2 epochs
Early-stopping metric	POS accuracy
Data splits	
Train / validation / test	70% / 15% / 15%
Random seed	42

Table B.1 Hyperparameters and training configuration for the multi-head morphological analyser.

Software and Hardware Environment

Training and evaluation were carried out in a Google Colab environment. Table B.2 lists the software and hardware used. Library versions correspond to the Colab default at the time of training.

Component	Version
Python	3.12.13
PyTorch	2.10.0+cu128
Transformers	5.0.0
Datasets	4.0.0
scikit-learn	1.6.1
NumPy	2.0.2
Matplotlib	3.10.0
Compute environment	Google Colab
GPU	single NVIDIA Tesla T4
CUDA	12.8

Table B.2 Software and hardware environment used for training and evaluating the morphological analyser.

Appendix C POS-Feature Applicability Table

A reference table for the POS to feature applicability matrix used by the morphological analysis model, as described in Section 3.4.1.

	person	number	tense	mood	voice	gender	case	degree
adjective	False	True	False	False	False	True	True	True
conjunction	False	False	False	False	False	False	False	False
adverb	False	False	False	False	False	False	False	True
relative adverb	False	False	False	False	False	False	False	False
interrogative adverb	False	False	False	False	False	False	False	False
foreign word	False	False	False	False	False	False	False	False
subjunction	False	False	False	False	False	False	False	False
interjection	False	False	False	False	False	False	False	False
cardinal numeral	False	True	False	False	False	True	True	False
ordinal numeral	False	True	False	False	False	True	True	False
infinitive marker	False	False	False	False	False	False	False	False
common noun	False	True	False	False	False	True	True	False
proper noun	False	True	False	False	False	True	True	False
reciprocal pronoun	False	True	False	False	False	True	True	False
demonstrative pronoun	False	True	False	False	False	True	True	False
interrogative pronoun	False	True	False	False	False	True	True	False
personal reflexive pronoun	True	True	False	False	False	True	True	False
personal pronoun	True	True	False	False	False	True	True	False
relative pronoun	False	True	False	False	False	True	True	False
possessive pronoun	True	True	False	False	False	True	True	False
possessive reflexive pronoun	True	True	False	False	False	True	True	False
indefinite pronoun	False	True	False	False	False	True	True	False
quantifier	False	False	False	False	False	False	False	False
preposition	False	False	False	False	False	False	False	False
article	False	False	False	False	False	False	False	False
verb	True	True	True	True	True	False	False	False
unassigned	False	False	False	False	False	False	False	False

Table C.1 POS to feature applicability table.

Appendix D Morphological Analyser Evaluation Results

This appendix contains the full per-class precision, recall, F1, and support for each feature head of the morphological analyser, evaluated on the PROIEL test set, as well as the diagrams produced during the model confidence evaluation.

D.1 Full Per-class Metric Results

Feature: POS				
Class	**Precision**	**Recall**	**F1**	**Support**
adjective	0.69	0.28	0.40	671
conjunction	0.97	0.97	0.97	1335
adverb	0.84	0.73	0.78	1472
relative adverb	0.91	0.27	0.42	74
interrogative adverb	0.71	0.59	0.64	66
foreign word	0.00	0.00	0.00	0
subjunction	0.90	0.87	0.88	517
interjection	0.91	0.28	0.43	74
cardinal numeral	0.85	0.49	0.62	158
ordinal numeral	0.80	0.23	0.36	35
common noun	0.60	0.75	0.67	3159
proper noun	0.59	0.36	0.45	629
reciprocal pronoun	0.00	0.00	0.00	1
demonstrative pronoun	0.92	0.61	0.73	500
interrogative pronoun	0.88	0.84	0.86	90
personal reflexive pronoun	1.00	0.98	0.99	42
personal pronoun	0.94	0.84	0.89	1101
relative pronoun	0.94	0.96	0.95	464
possessive pronoun	0.68	0.48	0.56	231
possessive reflexive pronoun	0.86	0.34	0.49	94
indefinite pronoun	0.82	0.29	0.43	294
preposition	0.97	0.93	0.95	1343
verb	0.69	0.85	0.76	4073
Macro Average	0.76	0.56	0.62	16423
Weighted Average	0.78	0.76	0.75	16423
Accuracy	0.76			16423

Table D.1 POS metric results.

Feature: PERSON				
Class	**Precision**	**Recall**	**F1**	**Support**
1	0.79	0.65	0.71	819
2	0.75	0.71	0.73	903
3	0.82	0.90	0.86	2842
None	0.74	0.67	0.70	977
Macro Average	0.77	0.73	0.75	5541
Weighted Average	0.79	0.79	0.79	5541
Accuracy	0.79			5541

Table D.2 Person metric results.

Feature: NUMBER				
Class	**Precision**	**Recall**	**F1**	**Support**
Plural	0.76	0.53	0.62	3496
Singular	0.80	0.93	0.86	7588
None	0.88	0.52	0.65	458
Macro Average	0.81	0.66	0.71	11542
Weighted Average	0.79	0.79	0.78	11542
Accuracy	0.79			11542

Table D.3 Number metric results.

Feature: TENSE				
Class	**Precision**	**Recall**	**F1**	**Support**
future	0.62	0.29	0.40	261
future perfect	0.51	0.33	0.40	66
imperfect	0.57	0.50	0.53	370
perfect	0.65	0.67	0.66	1153
pluperfect	0.63	0.52	0.57	94
present	0.78	0.85	0.81	2114
None	0.90	0.60	0.72	15
Macro Average	0.67	0.54	0.58	4073
Weighted Average	0.70	0.71	0.70	4073
Accuracy	0.71			4073

Table D.4 Tense metric results.

Feature: MOOD				
Class	**Precision**	**Recall**	**F1**	**Support**
gerundive	0.00	0.00	0.00	5
gerund	0.60	0.67	0.63	9
imperative	0.75	0.55	0.63	200
indicative	0.84	0.91	0.87	2452
infinitive	0.78	0.58	0.66	242
participle	0.73	0.66	0.69	720
subjunctive	0.75	0.72	0.73	444
supine	0.00	0.00	0.00	1
None	0.00	0.00	0.00	0
Macro Average	0.49	0.45	0.47	4073
Weighted Average	0.80	0.80	0.80	4073
Accuracy	0.80			4073

Table D.5 Mood metric results.

Feature: VOICE				
Class	**Precision**	**Recall**	**F1**	**Support**
Active	0.91	0.98	0.95	3485
Passive	0.79	0.44	0.56	573
None	0.82	0.60	0.69	15
Macro Average	0.84	0.67	0.73	4073
Weighted Average	0.90	0.90	0.89	4073
Accuracy	0.90			4073

Table D.6 Voice metric results.

Feature: GENDER				
Class	**Precision**	**Recall**	**F1**	**Support**
feminine	0.60	0.55	0.57	1686
feminine or neuter	0.00	0.00	0.00	2
masculine	0.73	0.89	0.80	4067
masculine or feminine	0.40	0.02	0.03	111
masculine or feminine or neuter	0.36	0.07	0.12	71
masculine or neuter	0.00	0.00	0.00	51
neuter	0.73	0.49	0.59	1275
None	0.89	0.49	0.63	206
Macro Average	0.46	0.31	0.34	7469
Weighted Average	0.69	0.71	0.69	7469
Accuracy	0.71			7469

Table D.7 Gender metric results.

Feature: CASE				
Class	**Precision**	**Recall**	**F1**	**Support**
ablative	0.87	0.75	0.81	1269
accusative	0.64	0.85	0.73	2333
dative	0.85	0.62	0.72	536
genitive	0.76	0.51	0.61	845
nominative	0.79	0.78	0.78	2187
vocative	0.70	0.51	0.59	93
None	0.88	0.48	0.62	206
Macro Average	0.78	0.64	0.69	7469
Weighted Average	0.76	0.74	0.74	7469
Accuracy	0.74			7469

Table D.8 Case metric results.

Feature: DEGREE				
Class	**Precision**	**Recall**	**F1**	**Support**
comparative	0.85	0.37	0.52	62
positive	0.82	0.83	0.83	681
superlative	0.00	0.00	0.00	13
None	0.91	0.94	0.92	1387
Macro Average	0.65	0.54	0.57	2143
Weighted Average	0.88	0.88	0.88	2143
Accuracy	0.88			2143

Table D.9 Degree metric results.

D.2 Confidence-Based Fallback

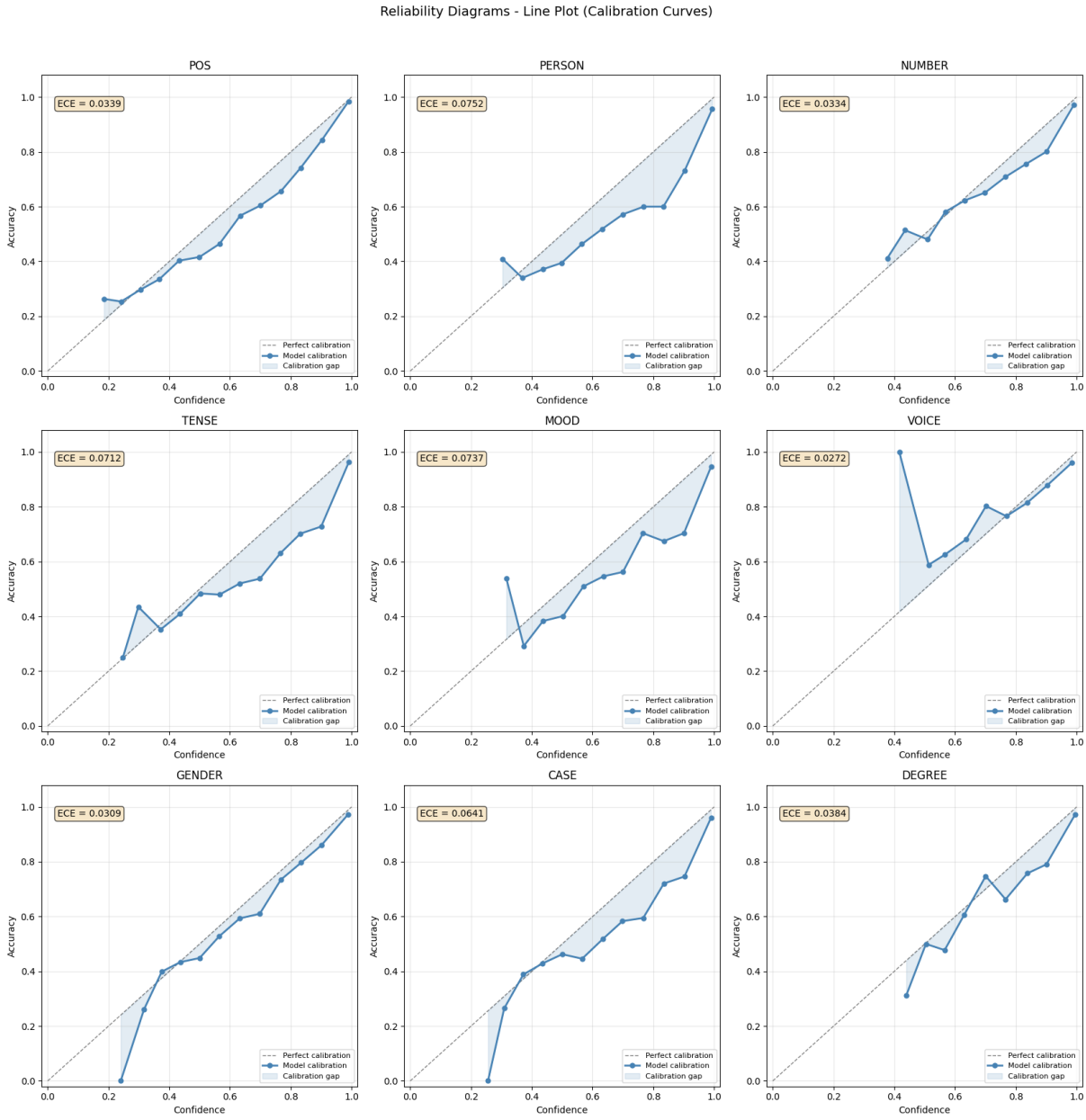


Figure D.1 Per-feature reliability diagrams on the validation set. Confidences are binned into 15 equal-width bins.

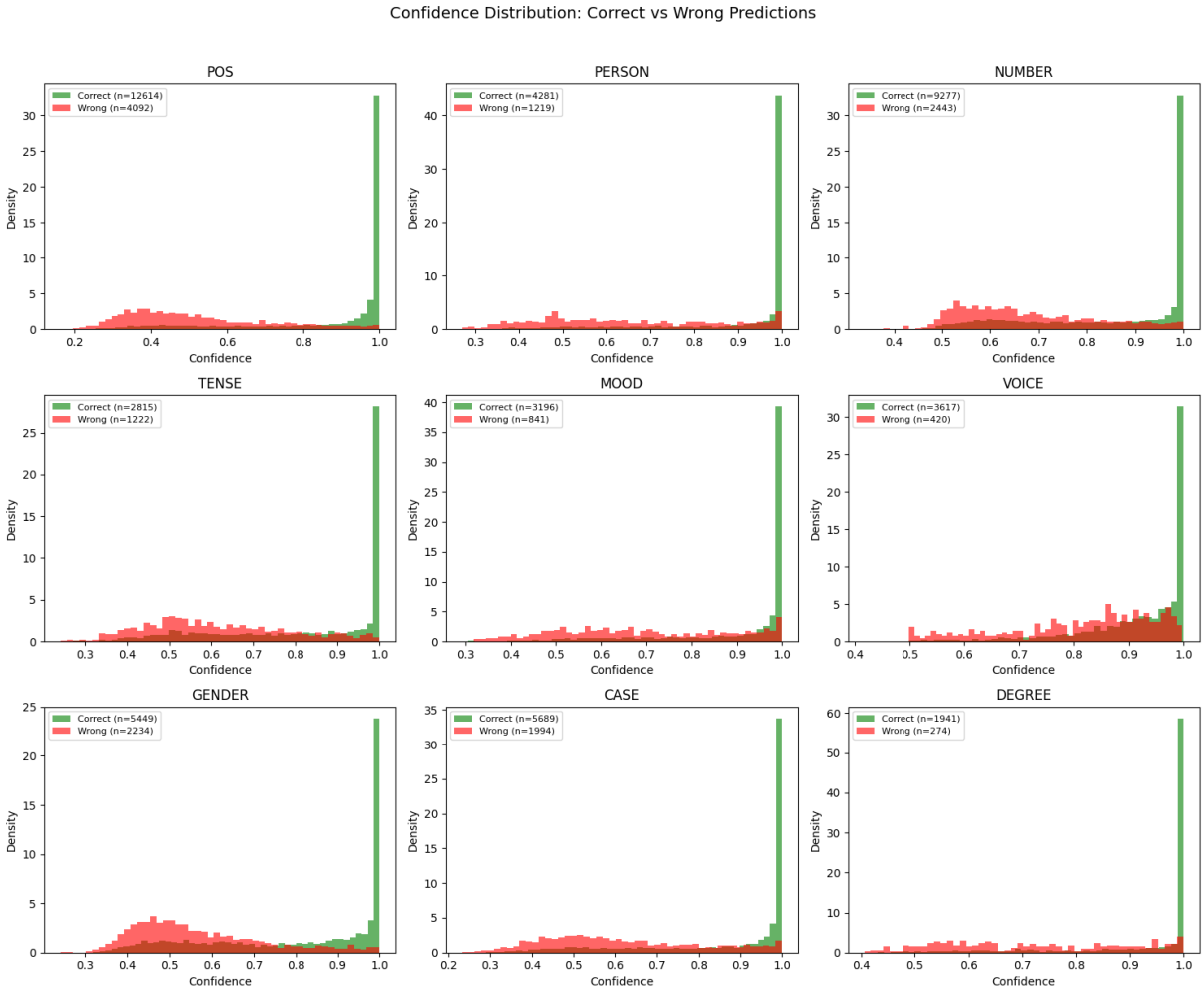


Figure D.2 Confidence distributions of correct (green) and incorrect (red) predictions per feature, normalised to densities.

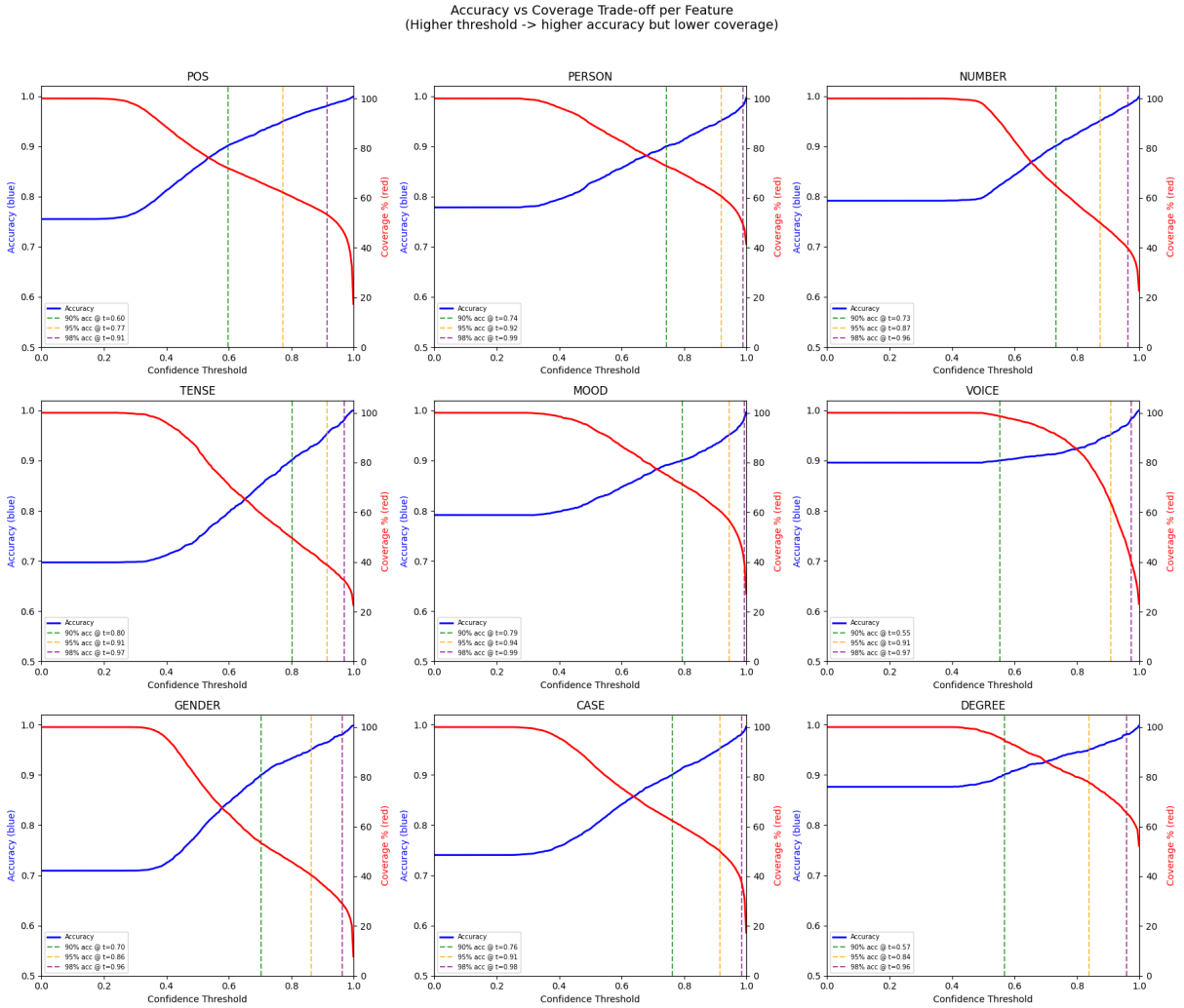


Figure D.3 Accuracy-coverage trade-off per feature.

D.3 Dataset Class Distributions

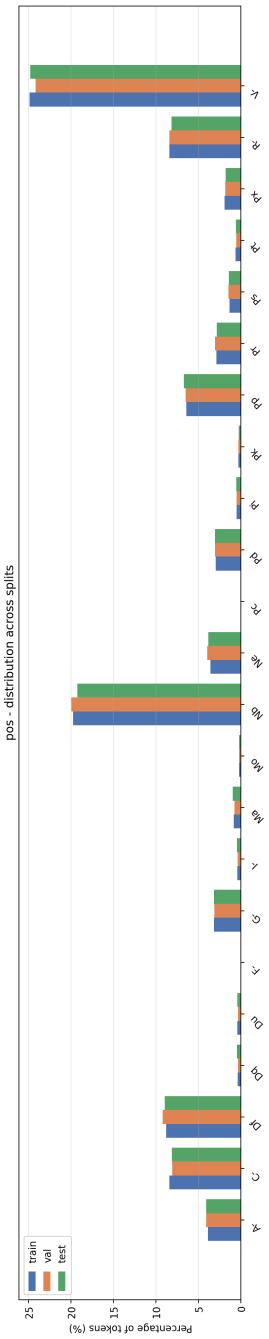
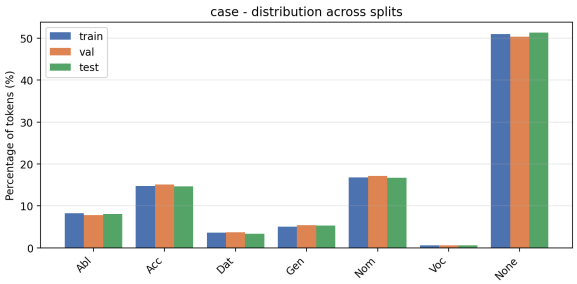
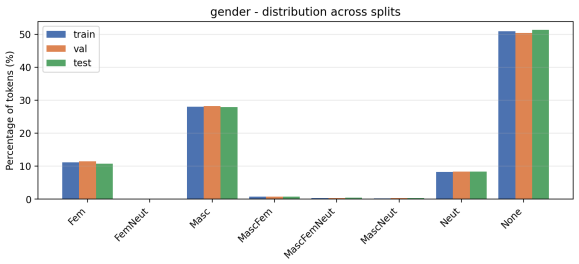


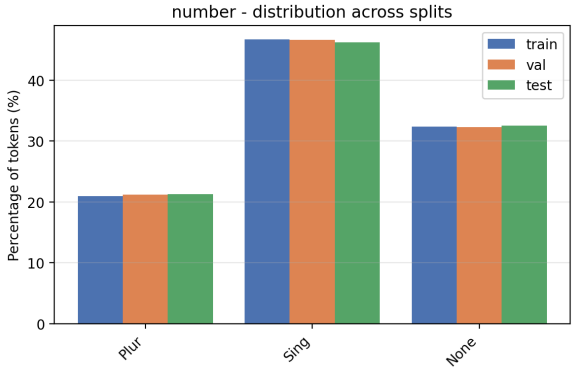
Figure D.4 Distribution of part-of-speech tags across train, validation, and test splits of the PROIEL Latin treebank.



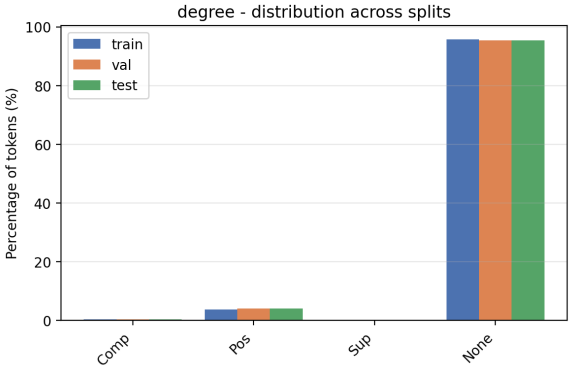
(a) Case



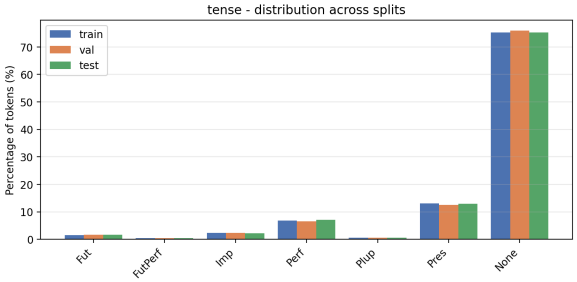
(b) Gender



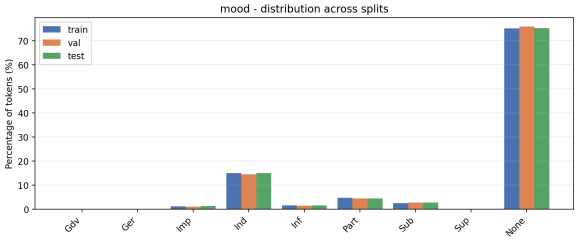
(c) Number



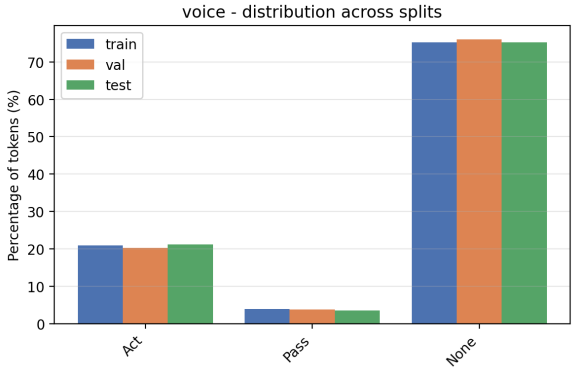
(d) Degree



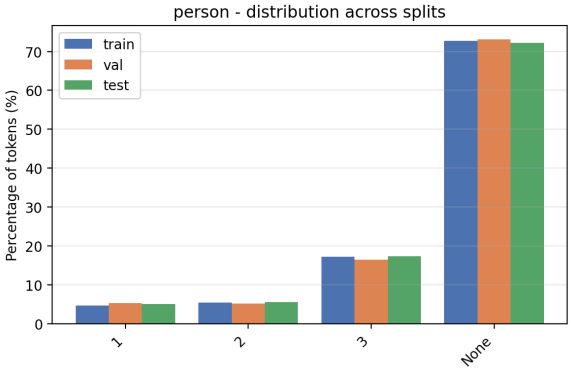
(e) Tense



(f) Mood



(g) Voice



(h) Person

Figure D.5 Distribution of morphological feature labels across train, validation, and test splits.

Appendix E User Evaluation Questionnaire

This appendix lists the items presented to participants in the user evaluation described in Section 4.4.

Background

- **How would you describe your level of Latin?**
  Single choice: No prior experience; Beginner; Intermediate; Advanced; Fluent / professional.
- **How often do you use AI chatbots (e.g. ChatGPT, Claude)?**
  Single choice: Never; Rarely; Sometimes; Often; Daily.

Likert Items

Participants responded on a 5-point scale: 1 = Strongly disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly agree.

1. The chatbot was easy to use and navigate.
2. The chatbot understood what I was asking.
3. The answers I received seemed accurate.
4. The answers I received were helpful for my learning.
5. I enjoyed using the chatbot to practice Latin.
6. Using the chatbot increased my interest in learning Latin.
7. I would recommend this tool to other Latin learners.

Open-ended Feedback

- What did you like most? (Optional)
- What could be improved? (Optional)
- Anything else you'd like to add? (Optional)